# Optimization of Maintenance Operations for Urban EV Charging Infrastructure under Uncertainty


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# Contents

## List of Figures



## List of Tables

# Abstract

Eine Kurzzusammenfassung der Vorgehensweise und der wesentlichen Ergebnisse.

Allgemeine Merkmale

- Objektivität: Es soll sich jeder persönlichen Wertung enthalten.
- Kürze: Es soll so kurz wie möglich sein.
- Verständlichkeit: Es weist eine klare, nachvollziehbare Sprache und Struktur auf.
- Vollständigkeit: Alle wesentlichen Sachverhalte sollen enthalten sein.
- Genauigkeit: Es soll genau die Inhalte und die Meinung der Originalarbeit wiedergeben.

# 1 Einleitung

# 2 Problem Definition

This chapter introduces the practical planning problem addressed in the thesis: the maintenance and disturbance handling of a network of electric vehicle charging stations.
The problem involves the daily routing and scheduling of mobile maintenance teams responsible for servicing a geographically distributed network of charging stations. The planning problem is not static: in addition to regular preventive tasks, new disturbance events may occur during the day and require operational responses on short notice. As a result, the system combines elements of preventive maintenance, corrective maintenance, and dynamic routing.
The decision process is organized on a daily basis. At the beginning of each day, an initial service plan is constructed for both teams. During the operating day, this plan may be revised in response to newly occurring faults.

## 2.1 Operational Setting

The operational scenario considered in this thesis is based on the charging infrastructure of the city of Würzburg, consisting of 397 stations distributed across the urban service area. These charging stations form the service network that must be maintained by two mobile service teams. Each team starts its route at a central depot and is required to return to the depot by the end of the working day. The service region is represented as a graph whose node set consists of the depot and all charging stations. Travel between stations is therefore a core element of the problem, since every maintenance action requires physical movement through the network.
The planning horizon is structured as a daily operational process. The working day runs from 8:00 to 17:00. At the beginning of the day, a full initial plan is generated. During the day, replanning decisions are triggered event-driven on an hourly grid whenever new disturbance events arrive. This means that routing decisions are not made once and then executed unchanged. Instead, they are revised as new information becomes available. Such a planning logic is essential in practice because maintenance operations in charging infrastructure are rarely fully predictable at the start of the day.
A further operational characteristic is the presence of a deterministic lunch break between 12:00 and 13:00. In the route-based setup, this break is not treated as an independent decision variable. Instead, it is inserted according to a fixed operational rule and shifts all subsequent arrival and departure times accordingly. This reflects the idea that some daily routines are known in advance and need not be optimized explicitly, but still influence route feasibility and time consumption.
The spatial structure of the system also plays an important role. Because the charging stations are distributed across the service region, route efficiency depends strongly on travel distances, travel times, and geographic clustering. For this reason, the operational logic uses zones as a practical pre-structuring device. The stations are grouped into zones which support the initial selection of promising service areas for the two teams. These zones re-

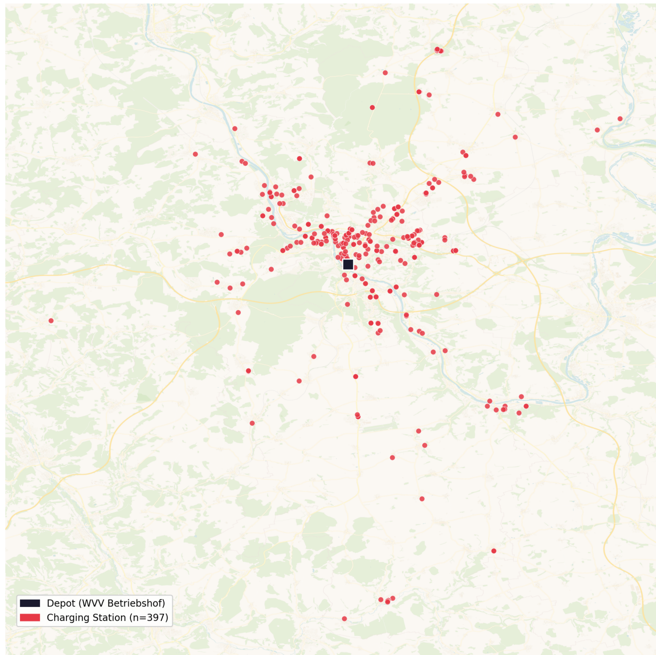


Figure 1: Spatial distribution of the 397 charging stations across the Würzburg urban service area, with the central depot marked.

duce routing complexity and guide the daily assignment of work areas, though the formal zone selection mechanism is introduced in detail in a later chapter.

Overall, the operational setting illustrates that the problem is a daily service planning task under operational constraints rather than a long-term static assignment problem. The interaction between limited team capacity, geographically dispersed stations, fixed working hours, and dynamically emerging disturbances creates a planning environment in which route design and real-time adaptation are equally important.

## 2.2 Environment and Disturbances

A defining feature of the planning problem is the presence of uncertainty. The maintenance teams operate in an environment in which service requirements are not fully known at the start of the day. New faults may occur while teams are already executing their planned routes, and the operational system therefore evolves over time. For this reason, a purely static routing approach is insufficient. Instead, the planning logic must account for the fact that the set of required service actions can change during execution.

In this setting, disturbances are treated as exogenous operational events. They occur stochastically and may emerge at different charging stations at different times of the day. Once a disturbance occurs, it immediately creates an additional service requirement that competes for limited maintenance capacity. This means that service planning must continuously balance already scheduled preventive work with newly arriving corrective tasks. The problem

is therefore not only about finding efficient routes, but also about deciding how to reallocate attention when the system state changes unexpectedly.
The route-based model distinguishes between two types of disturbances. Type-1 faults are simple failures that can be resolved through standard repair actions with a fixed service time. Type-2 faults are more complex and require disassembly, a depot roundtrip, and reassembly, resulting in a significantly longer and station-dependent service time. This distinction is operationally important because the insertion of a short Type-1 job and the insertion of a long Type-2 repair do not have the same impact on the remainder of the day.
Uncertainty is not limited to the occurrence of disturbances. Travel conditions also vary over time and are subject to uncertainty. Travel times depend on the time of day due to varying traffic conditions, and their actual realization is stochastic. This means that geographically similar routes may still differ substantially in execution quality depending on temporal traffic patterns and local congestion effects.
Another relevant feature of the environment is that unresolved disturbance jobs do not disappear at the end of the day. If a newly occurring fault cannot be inserted feasibly into the remaining route of either team, it is transferred as a carryover job to the following day. These carryover disturbances are then prioritized in the next daily planning cycle. This mechanism introduces an intertemporal dependency into the operational system: today's capacity shortage affects tomorrow's workload. In practical terms, the planning problem therefore includes not only immediate routing consequences but also backlog management over multiple days.

## 2.3 Assumptions and Cost Model

To make the planning problem analytically tractable, the operational system is represented through a number of simplifying assumptions. First, not every real-world process is modeled in full physical detail. Some side processes associated with complex repairs are incorporated through aggregate service-time components instead of explicit intermediate route legs. In the current formulation, a Type-2 fault includes additional time components for disassembly, depot-related handling, and reassembly, but the team remains conceptually assigned to the affected station rather than physically traveling to and from the depot as a separate route segment. This simplification reduces model complexity while preserving the main operational time impact of such events.
Second, service requirements are represented as operational jobs rather than as fully persistent technical condition states. Open regular maintenance tasks, new faults, and carryover disturbances are treated as actionable job sets. In addition, each charging station is associated with a maintenance-related status through the indicator of whether periodic maintenance has been completed and through the variable days since maintenance, which later drives failure risk. However, the binary distinction between "healthy" and "faulty" infrastructure is not carried as a separate persistent state variable in the operational problem description. This abstraction supports a cleaner transition into the route-based MDP in Chapter 4.
The economic evaluation of the planning problem is based on three cost components. The

first component consists of labor costs for the maintenance teams, set to 35 euros per hour. This figure is derived from an average gross wage of approximately 21 euros per hour for maintenance technicians in Bavaria (jobvector GmbH, 2024), scaled up by employer-side social security contributions of roughly 20–25 percent and an overhead allowance for provisions and downtime, resulting in a total employer cost of approximately 35 euros per hour. The second component consists of travel costs, set to 0.30 euros per kilometer, corresponding to the standard statutory mileage reimbursement rate for business travel in Germany (Bundesministerium der Finanzen, 2024). The third component consists of downtime-related costs, set to 0.50 euros per kilowatt-hour, approximating the market price for public AC charging in Germany as reported by the BDEW (Bundesverband der Energie- und Wasserwirtschaft, 2024). This rate reflects the opportunity cost of lost charging capacity per kilowatt-hour that could not be delivered due to an unresolved fault. Together, these three components reflect that good operational performance depends not only on efficient routing, but also on timely fault resolution.
The cost structure creates a clear trade-off. A route with low travel distance is not necessarily desirable if it leaves important faults unserved for too long. Conversely, a strongly service-oriented plan that aggressively inserts every disturbance may lead to high labor and travel expenditure or may jeopardize the timely return to the depot. The relevant operational objective is therefore not the minimization of a single physical quantity such as distance, but the minimization of total expected cost across labor use, mobility effort, and service delay consequences.

## 2.4 Motivating Example

To illustrate why both the initial plan and the replanning decision require more than routing efficiency, consider the simplified scenarios shown in Figures 2 and 3. The scenario involves

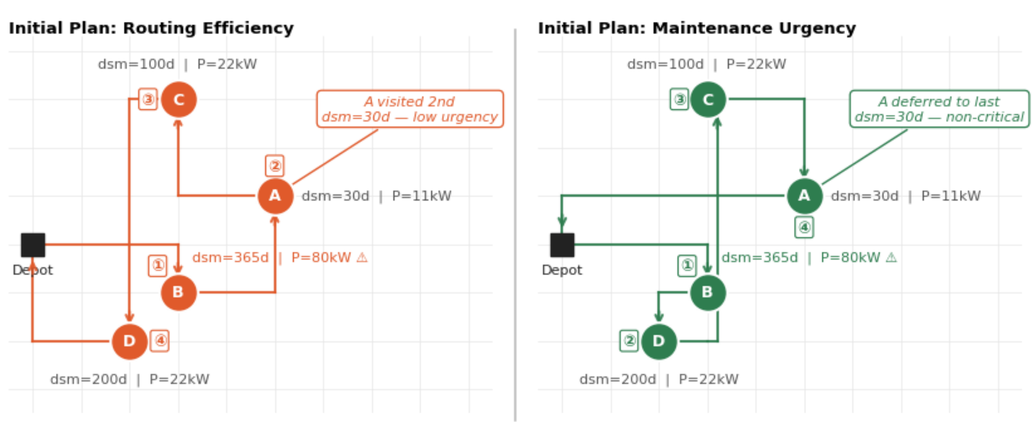


Figure 2: Two initial plan strategies for the same set of stations. Left: routing-efficiency approach visits A second despite low urgency. Right: maintenance-urgency approach defers A to last, ensuring critical stations are covered first.

four stations — A, B, C, and D — and a central depot. Station B has not been serviced in 365 days and operates at 80 kW, placing it at maximum failure risk. Station A was serviced only 30 days ago and carries only 11 kW — still non-critical. Stations C and D have intermediate maintenance states with 100 and 200 days since last service respectively.

Figure 2 contrasts two approaches to constructing the initial route. The left panel shows a routing-efficiency approach: stations are ordered primarily by geographic convenience, resulting in Depot $\rightarrow$ B $\rightarrow$ A $\rightarrow$ C $\rightarrow$ D. While B is correctly visited first, A is scheduled second despite its low urgency, consuming an early slot that could have served a more critical station. The right panel shows a maintenance-urgency approach: stations are ordered by their criticality, deferring A to the last position. The route becomes Depot $\rightarrow$ B $\rightarrow$ D $\rightarrow$ C $\rightarrow$ A. The key advantage becomes apparent when a disruption arrives later in the day and forces a drop: in the right panel, the critical stations B, D, and C have already been serviced, so A is the natural candidate for removal with negligible consequences. In the left panel, a disruption arriving after A might force the drop of C or D — stations with considerably higher maintenance urgency.

This shows that even the initial plan is a decision with future consequences: the order in which stations are visited determines which ones remain at risk if capacity is unexpectedly consumed by a disruption later in the day.

Figure 3 illustrates the replanning situation. Starting from the planned route, disturbance E occurs during execution and is inserted as the first stop after the depot. One station must be dropped to keep the route feasible. The left panel shows a purely distance-driven choice: B is dropped because it causes the largest routing detour, resulting in E $\rightarrow$ A $\rightarrow$ C $\rightarrow$ D. B remains unserviced, its days since maintenance increases to 366, and its failure risk reaches a critical level — a cost that will likely materialize the following day. The right panel shows a state-aware decision: A is dropped instead, as its low days since maintenance and low rated power make a one-day delay negligible. The route becomes E $\rightarrow$ B $\rightarrow$ D $\rightarrow$ C, B is visited, and its maintenance counter resets to zero, averting the failure risk entirely.

Both examples point to the same insight: good maintenance routing requires knowledge of each station's state — days since maintenance and rated power — not just geographic distance. A purely distance-driven approach ignores this information and can systematically produce worse outcomes. The central idea of the solution approaches developed in this thesis is therefore to assign an explicit value to each station based on its current maintenance state, and to use this value as a scoring criterion in both the initial plan construction and the replanning decision. This shifts the problem from a purely geometric routing task to a state-dependent optimization problem, in which the expected future cost of leaving a station unserved is weighed against the immediate routing detour required to serve it.

Chapter 3 reviews the relevant literature on maintenance optimization, stochastic vehicle routing, and approximate dynamic programming, providing the theoretical foundation for the solution approach. Building on both, Chapter 4 translates the operational setting into a route-based Markov decision process and connects that formulation to an approximate dynamic programming solution approach.

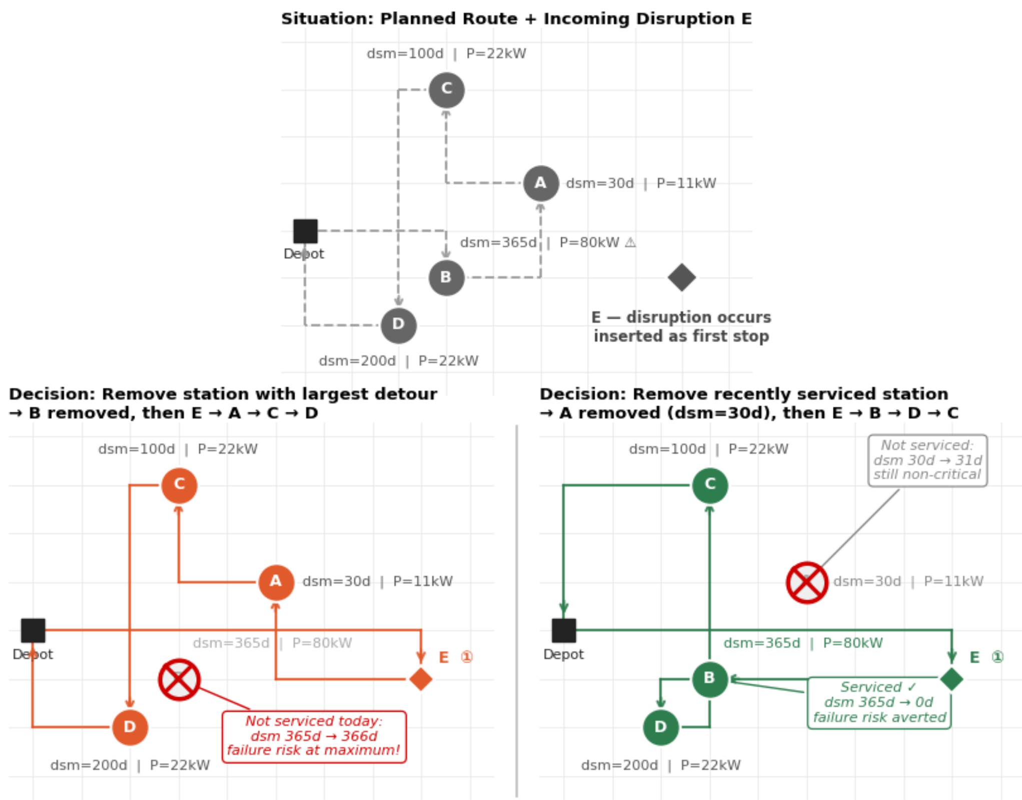


Figure 3: Two replanning decisions under the same incoming disruption E. Left: dropping B (largest detour) leaves the most critical station unserviced. Right: dropping A (recently serviced, low power) allows B to be serviced and averts failure risk.

# 3 Literature Review

The problem outlined in Chapter 2 sits at the intersection of three research streams: maintenance optimization, stochastic vehicle routing, and approximate dynamic programming. This chapter reviews each of these streams in turn and shows how they collectively provide the theoretical basis for the solution approach developed in this thesis.

## 3.1 Maintenance Optimization in Infrastructure Systems

Maintenance optimization in infrastructure systems addresses the challenge of planning, scheduling, and executing maintenance activities to ensure system availability, reliability, and cost-efficiency over the lifecycle of technical assets. Unlike manufacturing or logistics contexts, infrastructure maintenance often involves geographically distributed assets, limited maintenance resources, and significant consequences of system failures (de Jonge & Scarf, 2020; Dekker, 1996).

The following subsections trace this progression: from the core objectives and trade-offs of maintenance optimization, through the dominant maintenance strategies and their limits, to the specific complications that arise when technicians must travel between geographically distributed sites — the operational structure directly relevant to this thesis.

### 3.1.1 Concept and Objectives of Maintenance Optimization

Maintenance optimization refers to the systematic determination of when and what to maintain in order to achieve specific objectives such as minimizing costs, maximizing availability, or balancing both (de Jonge & Scarf, 2020). The fundamental trade-off lies between the cost of maintenance activities and the cost of system failures or downtime (Dekker, 1996).

Early work by Dekker (1996) identified six barriers preventing the application of maintenance optimization models in practice: models are difficult to understand, many papers are written only for mathematical purposes, maintenance activities consist of many different aspects, optimization is not always necessary, models often focus on the wrong type of maintenance, and there is a gap between theory and operational reality. Despite these challenges, maintenance optimization remains strategically important because continuous developments of technical systems and increasing reliance on equipment have resulted in growing importance of effective maintenance activities (de Jonge & Scarf, 2020).

In the context of charging infrastructure, maintenance optimization must address not only individual station failures but also fleet-level dispatching, resource allocation, and timing decisions under uncertainty (de Jonge & Scarf, 2020). The objective is typically to minimize total expected costs (maintenance plus failure costs) while maintaining a required level of system availability (Alaswad & Xiang, 2017; Dekker, 1996).

### 3.1.2 Maintenance Strategies: Preventive, Reactive, Condition-Based

Maintenance strategies can be broadly classified into three categories: reactive (corrective) maintenance, preventive (planned) maintenance, and condition-based (predictive) maintenance (Sarja & Halilov, 2023; de Jonge & Scarf, 2020).

Reactive maintenance (also called corrective or run-to-failure maintenance) is performed only after a failure occurs (Sarja & Halilov, 2023). This strategy is suitable for non-critical assets where downtime has minimal impact on operations. However, for critical infrastructure such as charging stations, reactive maintenance can lead to significant customer dissatisfaction, safety risks, and higher repair costs due to unanticipated failures (de Jonge & Scarf, 2020).

Preventive maintenance is performed according to a fixed schedule or usage-based interval, regardless of the actual condition of the asset (Sarja & Halilov, 2023; Alaswad & Xiang, 2017). Preventive maintenance aims to prevent failures before they occur by replacing or servicing components at predetermined times. This approach is well-established and widely applied, for example in vehicle maintenance or periodic inspections of technical systems (Dekker, 1996). However, preventive maintenance can lead to over-maintenance (replacing components that are still functional) or under-maintenance (failures occurring before the scheduled interval) (de Jonge & Scarf, 2020).

Condition-based maintenance (CBM) performs maintenance based on the monitored condition of the asset, for example sensor data or error rates, and is often extended to predictive maintenance using machine learning or digital twins (Alaswad & Xiang, 2017; Sarja & Halilov, 2023). While CBM is well-studied in the literature, it requires continuous condition monitoring infrastructure that is not assumed in this thesis.

In practice, many organizations use a hybrid strategy, combining preventive maintenance for critical components with reactive maintenance for non-critical ones and condition-based maintenance for assets where monitoring is available (Sarja & Halilov, 2023; de Jonge & Scarf, 2020). The charging infrastructure context falls naturally into this hybrid territory: routine inspections address predictable deterioration, while spontaneous hardware failures demand immediate reactive response — a combination that motivates the dynamic replanning logic developed in Chapter 5.

### 3.1.3 Uncertainty, Availability, and Cost as Central Decision Variables

Maintenance decisions are inherently made under uncertainty. Key uncertain parameters include failure times, service times, travel times, availability of spare parts, and arrival of new maintenance requests (de Jonge & Scarf, 2020; Dekker, 1996). These three dimensions — availability, cost, and uncertainty — jointly define the scope of maintenance optimization:

- **Availability** measures the proportion of time that a system is operational and able to fulfill its function (Sarja & Halilov, 2023). For charging infrastructure, failed stations directly reduce customer access and revenue, making availability a primary performance indicator.
- **Cost** encompasses maintenance expenditures (labor, materials, travel), failure costs (downtime, customer dissatisfaction, penalties), and inventory costs (spare parts, tools) (de Jonge & Scarf, 2020; Dekker, 1996). The optimization objective is typically to minimize total expected cost over a planning horizon, subject to availability constraints (Alaswad & Xiang, 2017).
- **Uncertainty** affects both the timing and the magnitude of maintenance consequences. A failure may occur earlier or later than expected, service may take longer than planned, or travel to a site may be delayed by traffic (de Jonge & Scarf, 2020). Decision-making therefore follows a sequential process in which uncertainty gradually resolves and decisions are revised in response (de Jonge & Scarf, 2020).

In maintenance optimization, uncertainty is typically modeled through stochastic deterioration processes, probability distributions for failure times, or scenario-based approaches, with the choice depending on data availability, system complexity, and computational requirements (Alaswad & Xiang, 2017; de Jonge & Scarf, 2020).

### 3.1.4 Transition to Mobile Maintenance and Service Interventions

The maintenance problems discussed in Sections 3.1.1–3.1.3 typically focus on when and what to maintain at individual assets. However, in many practical settings, maintenance is performed by mobile technicians who must travel between geographically distributed sites,

which introduces routing and scheduling decisions (de Jonge & Scarf, 2020).
For charging infrastructure maintenance, this means that the optimization problem is not only about determining maintenance intervals or failure thresholds but also about dispatching technicians, planning routes, and allocating limited resources (vehicles, spare parts, time) across multiple stations (de Jonge & Scarf, 2020; Alaswad & Xiang, 2017).
This transition from single-asset maintenance optimization to mobile maintenance service planning introduces additional complexity:

- **Spatial dependency:** Maintenance decisions for one station affect the availability of resources for other stations (de Jonge & Scarf, 2020).
- **Temporal dependency:** The order in which stations are visited affects total travel time and service availability (Alaswad & Xiang, 2017).
- **Uncertainty coupling:** Failures, travel times, and service times are often stochastic and interdependent (de Jonge & Scarf, 2020).

The literature on maintenance optimization has traditionally focused on single-asset or multi-asset models without explicit routing (de Jonge & Scarf, 2020; Dekker, 1996). More recent work begins to integrate maintenance planning with vehicle routing, especially in the context of dynamic and stochastic vehicle routing problems (SDVRP) (de Jonge & Scarf, 2020). This integration is precisely the focus of the following Sections 3.2 and 3.3, which cover stochastic routing and approximate dynamic programming for dynamic service planning.

### 3.1.5 EV Charging Infrastructure as a Maintenance Context

The maintenance of electric vehicle charging infrastructure constitutes an emerging and practically relevant application within the broader field of infrastructure maintenance optimization. As EV adoption accelerates globally, the scale and geographic dispersion of charging networks have grown substantially, making systematic maintenance planning a critical operational challenge for charging spot operators (Zhang, Sun, Jiang, & Cui, 2024).
EV charging stations exhibit two distinct operational requirements that together define the maintenance workload. First, charging piles require frequent inspections to identify potential safety hazards such as overheating, communication failures, or display errors. While individual inspections are relatively simple, the sheer number of stations makes inspection routing a non-trivial planning problem. Second, a substantial share of stations are subject to outright failures: empirical data from Chinese EV charging operators indicate that more than ten percent of deployed charging spots contain faulty charging piles at any given time, necessitating corrective maintenance by qualified engineers (Zhang et al., 2024). Taken together, operations and maintenance costs account for approximately 30–50 % of the total cost of running a charging spot network, making efficient maintenance routing a significant economic lever (Zhang et al., 2024).
Despite this practical relevance, the academic literature on EV charging station maintenance routing remains thin. The closest related work is Zhang et al. (2024), who propose a bi-objective routing optimization model for the large-scale inspection-maintenance of EV

charging spots in Xi'an, China, covering 1,281 stations. Their multi-stage approach applies K-Means clustering to reduce problem scale, estimates cluster-level costs via a greedy TSP heuristic, and solves the reduced model with an adaptive NSGA-II to obtain a Pareto frontier balancing total cost and composite customer satisfaction loss. This is, to the best of the author's knowledge, one of the first papers to explicitly address routing optimization for EV charging station maintenance.

However, the model of Zhang et al. (2024) is static and deterministic: all maintenance and inspection tasks are planned once at the beginning of the horizon, with no mechanism for intraday disruptions, stochastic fault arrivals, or replanning. In practice, charging stations fail unpredictably during the working day, and maintenance teams must revise their routes in response. This dynamic dimension — recurring daily over a multi-month operational horizon, with state-dependent task priorities that change as stations age between visits — is precisely what the present thesis addresses. The contribution of this work therefore lies not in replacing static routing models, but in extending the planning paradigm toward a dynamic, state-aware, and intertemporally consistent maintenance policy for EV charging infrastructure.

## 3.2 Stochastic Vehicle Routing Problems and Dynamic Service Planning

Efficient service planning in real-world environments is rarely a one-time optimization problem. Instead, it involves a sequence of decisions under uncertainty, where new information becomes available over time and must be incorporated into the planning process. This is particularly true for maintenance operations on charging infrastructure, where breakdowns, service requests, and priority changes can emerge unpredictably during the day.

The literature on Vehicle Routing Problems (VRP) has evolved from static, deterministic formulations toward dynamic and stochastic models that better reflect such operational realities (Ritzinger, Puchinger, & Hartl, 2016; Mardešić, Erdelić, Mardešić, & Đurasević, 2023). This section provides the theoretical foundation for understanding how stochastic and dynamic routing problems are modeled and solved.

### 3.2.1 Basic Concepts of VRP and DVRP

The classical Vehicle Routing Problem (VRP) involves planning optimal routes for a fleet of vehicles to serve a set of customers under various operational constraints (Mardešić et al., 2023; Toth & Vigo, 2014, as cited in Ritzinger et al., 2016). Common constraints include vehicle capacity and service time windows (Hildebrandt et al., 2023). In the static VRP, all relevant information is known at the beginning and does not change during planning (Pillac, Gendreau, Guéret, & Medaglia, 2013; Ritzinger et al., 2016). The dynamic Vehicle Routing Problem (DVRP) extends this basic model by incorporating information that emerges over time, for example new requests, changing travel times, or additional service needs (Abbatecola, Fanti, & Ukovich, 2016; Ritzinger et al., 2016). The stochastic Vehicle Routing Problem (SVRP) further augments this dynamism by explicitly modeling uncertainty, i.e., random or only partially known events (Ritzinger et al., 2016; Akhmetbek, 2023). For the literature, it is important that "dynamic" and "stochastic" often occur together but are not identical: a problem can be dynamic without being strongly stochastic, or stochastic without being re-planned in real time (Ritzinger et al., 2016). The survey literature shows that research has increasingly moved from the static VRP toward dynamic and stochastic variants because real logistics and service processes are rarely fully planable (Ritzinger et al., 2016; Mardešić, Erdelić, Mardešić, & Đurasević, 2023). For this thesis, this transition is central because maintenance tasks for charging infrastructure cannot be planned completely deterministically either (Ulmer, Goodson, Mattfeld, & Hennig, 2019; Ritzinger et al., 2016).

### 3.2.2 Uncertainties in Routing: Requests, Service Times, Travel Times, Capacities

A typical uncertainty aspect in the Stochastic Dynamic Vehicle Routing Problem (SDVRP) is the arrival of new customer requests or service requests that occur only during the tour (Zou & Dessouky, 2018; Ritzinger et al., 2016). While Ulmer et al. (2019) focus on general service requests like roadside assistance, this thesis applies this logic to maintenance orders and spontaneous failures at charging infrastructure. Another uncertainty factor is service times, which are hardly predictable exactly in practice because repairs may take different amounts of time or technical problems may occur on site (Ritzinger et al., 2016; Akhmetbek, 2023). Travel times can also be stochastic, for example due to traffic, weather, roadworks, or time-dependent speeds (Mardešić et al., 2023; Ritzinger et al., 2016). Additionally, capacities can be uncertain, for example when vehicles carry only limited tools, spare parts, charging energy, or personnel resources (Basso et al., 2022; Akhmetbek, 2023). The literature usually distinguishes several types of uncertainty, including stochastic demand, stochastic travel times, and stochastic arrival times (Ritzinger et al., 2016; Ulmer, Köster, & Mattfeld, 2018). For modeling, it is important which uncertainties are known ex ante and which are observed only online during the tour (Zou & Dessouky, 2018; Ulmer et al., 2019). Particularly relevant for this work is the fact that uncertainty affects not only the optimal sequence of stops but also whether an intervention is worthwhile at all or whether one should wait for further information (Ulmer et al., 2019; Ritzinger et al., 2016). It is precisely at this point that the transition from classical routing to dynamic service planning becomes visible (Mardešić et al., 2023; Ulmer et al., 2019).

### 3.2.3 Modeling Approaches: MDP, Multi-Stage Stochastic Programming, Look-ahead

Many works in the research matrix model the SDVRP as a Markov Decision Process (MDP) because decisions must be made sequentially under uncertainty (Ulmer et al., 2019; Mardešić et al., 2023). In an MDP, the problem is described by states, actions, transition probabilities, and an objective function (Ulmer et al., 2019). The state can contain vehicle positions, remaining time, open requests, priorities, or current loads (Ulmer et al., 2019; Ritzinger et al., 2016). An action is then, for example, the choice of the next intervention site, waiting for new information, or the decision whether to accept a job directly (Ulmer et al., 2019). The major advantage of this modeling is that it explicitly represents the dynamics of the problem and incorporates future uncertainty into the decision (Ulmer et al., 2019; Mardešić et al., 2023). However, the literature also emphasizes that an exact MDP quickly becomes very large because the number of possible states and actions explodes (Ulmer et al., 2019; Ritzinger et al., 2016). Therefore, approximate approaches such as Approximate Dynamic Programming (ADP) or look-ahead-based procedures are frequently used in practice (Ulmer et al., 2019; Ulmer, 2020). Alternatively, some works employ multi-stage stochastic programming models, where multiple decision stages with uncertainty realizations are optimized (Ritzinger et al., 2016; Mardešić et al., 2023). These models are conceptually elegant but often difficult to solve for large, realistic instances (Ritzinger et al., 2016). Look-ahead procedures, by contrast, consider only a limited planning horizon and attempt to make the short-term best decision while taking some future scenarios into account (Zou & Dessouky, 2018; Ulmer et al., 2019). For this thesis, this modeling family is particularly relevant because it lies closer to real maintenance and dispatching processes than purely static tour planning (Ulmer et al., 2019; Ulmer, 2020). The central insight from the literature is: the more uncertainty and real-time dynamics are embedded in the problem, the more one needs MDP- or look-ahead-based formulations instead of classical routing models (Ulmer et al., 2019; Mardešić et al., 2023).

### 3.2.4 Solution Families: Exact Methods, Heuristics, Metaheuristics

The literature on SDVRP shows several solution families that differ primarily in accuracy, scalability, and online capability (Ritzinger et al., 2016; Mardešić et al., 2023). Exact methods attempt to solve the problem mathematically optimally, for example via Mixed-Integer Programming (MIP), dynamic programming, or exact search procedures (Pillac et al., 2013; Ritzinger et al., 2016). These approaches are useful for small instances but quickly reach limits with dynamic, stochastic, and large-scale problems (Ritzinger et al., 2016; Akhmetbek, 2023). Therefore, heuristic procedures dominate the SDVRP literature, i.e., rules or search procedures that deliver good but not necessarily optimal solutions (Ritzinger et al., 2016; Mardešić et al., 2023). Typical heuristics include myopic rules, greedy dispatching, and insertion heuristics, which are particularly attractive in dynamic settings because they allow fast online decisions (Pillac et al., 2013; Sarasola et al., 2016). Metaheuristics such as Variable Neighborhood Search or Tabu Search can improve solution quality but tend to handle future uncertainty less explicitly than ADP-based approaches, making them less suitable as the primary method in anticipative maintenance planning (Ulmer et al., 2019; Ritzinger et al., 2016). For this thesis, greedy insertion serves as the operational routing primitive within all developed policies, while the anticipative element is introduced through the ADP layer described in Section 3.3.

### 3.2.5 Route-based MDPs and Their Relevance for This Thesis

Route-based MDPs are particularly important because they model the routing problem at the level of the route rather than only at the level of individual states or customer requests. Ulmer et al. (2020) formally introduce this framework, defining route-based MDPs as a generalization of conventional MDPs in which the action space, state space, and reward structure are augmented to operate on routing plans rather than individual node selections. The advantage lies in the fact that the route itself is understood as a decision unit, and not only individual stops are considered in isolation. This allows anticipative decisions, i.e., decisions that do not only react to the current state but consider expected future events (Ulmer, 2020; Mardešić et al., 2023). Particularly relevant are the works on Approximate Dynamic Programming, offline-online ADP, and horizontal combinations of online and offline methods (Ulmer, 2020; Ulmer et al., 2019). These works show that route-based MDPs can be well combined with ADP strategies because the value function or rollout logic can approximate future developments (Ulmer et al., 2019; Ulmer, 2020). For this thesis, this is a strong argument because maintenance interventions also do not consist only of individual decisions but of a sequence of interdependent route and intervention decisions (Ulmer et al., 2019). Particularly fitting is the idea that failures or service requirements occur during the day and therefore are not fully known in advance (Ulmer et al., 2019; Ritzinger et al., 2016). The route-based perspective allows exactly this kind of dynamic, sequential, and resource-aware planning (Ulmer, 2020; Ulmer et al., 2019). Route-based MDPs therefore serve as the methodological hinge between the SDVRP literature reviewed in this section and the ADP framework introduced in Section 3.3. They provide the formal structure within which anticipative decision-making for charging station maintenance can be both modeled and approximated (Ulmer et al., 2019; Mardešić et al., 2023).

### 3.2.6 Limits of Classical SDVRP Models for Maintenance Tasks

Classical SDVRP models often originate from urban logistics, last-mile delivery, or courier-service contexts (Mardešić et al., 2023; Ritzinger et al., 2016). These applications are structurally similar but not fully transferable to maintenance logistics (Abbatecola et al., 2016; Akhmetbek, 2023). In maintenance scenarios, different decision priorities arise, such as technical criticality, failure consequences, recommissioning, and personnel qualification (Mardešić et al., 2023). Furthermore, maintenance interventions are often less standardized than deliveries because diagnosis, repair, and feedback can vary greatly (Abbatecola et al., 2016; Mardešić et al., 2023). Many SDVRP models treat requests as customer orders, whereas for this thesis, failure or maintenance orders are more relevant (Ulmer et al., 2019; Akhmetbek, 2023). This means that not only the spatial sequence but also technical prioritization and temporal urgency play a role (Ulmer et al., 2019; Mardešić et al., 2023). Another difference is that for charging infrastructure maintenance, return trips, spare parts, technical rework, and multi-visit scenarios can occur frequently (Abbatecola et al., 2016; Mardešić et al., 2023). Many literature models abstract these aspects strongly and are therefore only conditionally realistic (Ritzinger et al., 2016; Akhmetbek, 2023). The strength of the SDVRP literature thus lies less in a direct template for this problem but rather in transferable decision principles (Ritzinger et al., 2016; Mardešić et al., 2023). There are strong methods for dynamic routing problems, but the specific maintenance context of charging infrastructure remains underrepresented. This gap motivates the approach taken in this thesis: adopting the structural logic of route-based SDVRPs while adapting the priority rules, cost model, and replanning logic to the requirements of technical maintenance operations (Mardešić et al., 2023; Ulmer et al., 2019).

#### 3.2.7 Spatial Clustering as a Pre-Structuring Mechanism

A recurring approach in EV charging maintenance routing is the use of spatial clustering to decompose a large station network into manageable service zones before applying route optimization. Zhang et al. (2024) apply K-Means clustering to a network of 1,281 EV charging stations in Xi'an, China, where clustering is used to reduce the combinatorial scale of the problem: maintenance stations are retained individually while inspection spots are grouped into clusters, and the resulting reduced problem is then solved with an adaptive NSGA-II metaheuristic to minimize total cost and customer satisfaction loss simultaneously. Their multi-stage approach demonstrates that geographic pre-structuring is a practically necessary step when the network scale would otherwise make direct optimization intractable. Yenigun et al. (2025) adopt a similar spatial decomposition for 100 stations in Istanbul, grouping stations via K-Means and subsequently optimizing inter-cluster routing with a genetic algorithm, achieving a 35 % reduction in total travel distance compared to a sequential baseline. Both works share the core idea with this thesis that K-Means clustering over station coordinates provides a natural and computationally efficient way to organize service zones. However, a key difference is that both papers treat clustering as a one-time, static pre-processing step applied to the full station network before routing begins. In this thesis, by contrast, zone membership is fixed but zone *selection* is a daily decision: at the start of each working day, the system evaluates all open zones and assigns teams to those with the highest current priority, taking into account backlog, station criticality, and team positions. This repeated, state-dependent zone selection introduces a dynamic layer that static clustering approaches do not address and constitutes one of the structural contributions of the thesis.

## 3.3 Approximate Dynamic Programming for Dynamic Routing Problems

The dynamic and stochastic nature of real-world service planning requires decision frameworks that can handle sequential choices under uncertainty without prohibitive computational costs. Exact Dynamic Programming (DP) provides a theoretically optimal foundation but quickly becomes intractable for realistic routing problems. Approximate Dynamic Programming (ADP) addresses this limitation by approximating future costs or values rather than computing them exactly, thereby enabling scalable, anticipative decision-making in operational settings. This section introduces the core concepts, methods, and relevance of ADP for dynamic routing.

#### 3.3.1 Motivation: Why Exact Dynamic Programming Does Not Scale

Exact Dynamic Programming is theoretically elegant for routing problems with uncertainty but often practically intractable because the state space explodes rapidly. Even with moderate fleet sizes, multiple open requests, and dynamic events, an enormous combinatorial structure arises (Ulmer, 2017; Ritzinger, Puchinger, & Hartl, 2016). The literature consistently shows that exact DP serves as a reference but is too computationally intensive for real-world SDVRP instances (Ulmer et al., 2019; Pillac, Gendreau, Guéret, & Medaglia, 2013). This is precisely where ADP begins: instead of computing the exact optimal solution, ADP approximates future costs or future utility well enough to make good decisions. While Ulmer et al. (2018) focus on the time-sensitive nature of parcel pickup services, their logic regarding the need for fast and robust decisions is directly applicable to the maintenance context. In both domains, the dispatcher cannot wait for mathematical perfection but must react immediately to stochastic events – such as spontaneous breakdowns – to ensure operational capability. The ADP literature thereby justifies the transition from theoretical optimality to operational decision-making power (Mardešić et al., 2023). Particularly the Ulmer works show that routing problems with stochastic requests can realistically only be solved online with approximate procedures (Ulmer, 2017; Ulmer et al., 2019). Exact DP remains valuable as a conceptual benchmark, but ADP provides the practical framework for real-time service planning under uncertainty.

### 3.3.2 Basic Idea of ADP: States, Actions, Value Functions

ADP treats the problem as a sequential decision under uncertainty. A state describes the current planning status, for example vehicle positions, remaining time, open orders, the route traveled so far, or available resources (Ulmer, 2017; Ulmer et al., 2019). An action is a concrete decision in the current state, for example which order to serve next or whether to wait for new information (Ulmer, 2017). The value function describes how good a state or a state-action combination is in the long run. The central point is that future costs or returns are not computed exactly but approximated (Ulmer, 2017; Hildebrandt, Thomas, & Ulmer, 2023). This approximation can happen via linear functions, features, simulations, or data-driven approximations (Ulmer, 2020; Ulmer et al., 2019). That is precisely why ADP is more flexible than exact Dynamic Programming approaches. Ulmer (2017) and Ulmer et al. (2019) apply this logic to evaluate routing decisions with expected future quality. For this work, this way of thinking is useful because maintenance interventions also start from a current state, and the quality of a decision only shows itself over the future. In mathematical terms, ADP approximates the Bellman equation:

$$V_t(s_t) \approx \mathbb{E}\left[\sum_{k=t}^{T} \gamma^{k-t}\, C_k(s_k, a_k) \;\middle|\; s_t\right]$$

where $V_t(s_t)$ is the approximated value of state $s_t$ at time $t$, $C_k$ is the cost at stage $k$, $a_k$ is the action, and $\gamma$ is a discount factor (Ulmer, 2017; Bertsekas, 2017). Instead of computing this expectation exactly, ADP uses approximation functions learned offline or updated online.

### 3.3.3 Important ADP Approaches: Rollout, VFA, and Offline-Online ADP

Rollout is one of the most practically important ADP heuristics. A base policy is used as a reference, and for possible actions an anticipative future simulation is performed (Ulmer et al., 2019; Zou & Dessouky, 2018). Rollout is particularly strong when one must decide quickly online but still wants to account for the future at least approximately (Ulmer, 2019; Zhang, Ohlmann, & Thomas, 2018). Value Function Approximation (VFA) replaces the exact value function with an approximated function. The approximation can be based on features, partitionings, or historical data (Ulmer, 2020; Ulmer et al., 2019). In the research matrix, VFA is mentioned multiple times as a dominant ADP component, especially in the review works and in the offline-online approaches (Mardešić et al., 2023; Ulmer, 2020). Offline-Online ADP combines an offline learned or precomputed value function with an online executed decision rule (Ulmer et al., 2019). This is particularly attractive because computational effort is shifted into the preparation phase while remaining responsive online (Ulmer, 2019; Ulmer, 2020). As Ulmer et al. (2019, p. 185) note:

> The offline-online policy enhances the anticipation of the VFA policy, yielding spatial and temporal anticipation of requests and routing developments. The combination of VFA with rollout algorithms demonstrates the potential benefit of using offline and online methods in tandem as a hybrid ADP procedure, making possible higher-quality policies with reduced computational requirements for real-time decision making.

A related concept worth noting is the post-decision state: the system state immediately after an action is taken but before random information is revealed (Ulmer, 2017). This separation of deterministic action effects from subsequent stochasticity is conceptually useful, and the rollout-based policies developed in Chapter 5 implicitly rely on it — each Monte Carlo scenario begins from the post-decision state after a drop or insertion choice, before new faults are drawn. For this thesis, Rollout, VFA, and Offline-Online ADP are the three central paradigms. Their combination — learned value weights applied offline, rollout evaluation applied online — is precisely the structure realized in the solution policies of Chapter 5 (Ulmer, 2020; Ulmer et al., 2019).

### 3.3.4 Distinction from Reinforcement Learning

Reinforcement Learning (RL) and ADP share the sequential decision framework but differ in origin: RL emphasizes learning from interaction data, while ADP operates from the Operations Research tradition using explicit model-based structure (Hildebrandt et al., 2023; Mardešić et al., 2023). For complex route-based problems, ADP tends to be more robust because it separates search and evaluation explicitly via value functions or rollout logic, whereas many RL approaches scale poorly or fail to capture routing combinatorics (Hildebrandt et al., 2023; Joe & Lau, 2020). This thesis follows the ADP paradigm — using simulation-based rollout and learned value weights rather than end-to-end RL policy training.

### 3.3.5 Relevance of ADP for Dynamic Maintenance Decisions

ADP is particularly suitable for maintenance decisions because maintenance is typically not decided once but sequentially and under uncertainty (Ulmer, 2017; Ulmer et al., 2019). A defective charger, an unplanned failure, or a changing priority are exactly the type of events that ADP can address (Ulmer, Köster, & Mattfeld, 2018). The future does not need to be known exactly; it is sufficient to approximate it plausibly and align the decision with it (Ulmer, 2020; Ulmer et al., 2019). Offline-Online ADP is particularly attractive when maintenance rules can be trained or precomputed in advance and then applied quickly in operation (Ulmer, 2019; Ulmer, 2020). Rollout is sensible when on-site decisions must be made between several interventions or locations and the consequences of options should be approximately simulated (Zou & Dessouky, 2018; Zhang et al., 2018). VFA is useful when one wants to build a robust evaluation of states from experience data or simulated scenarios (Ulmer, 2020; Ulmer et al., 2019). Post-decision States are useful because they cleanly separate the effect of an action from the subsequent entry of uncertainty (Anuar et al., 2022; Ulmer, 2017). Overall, the literature shows that ADP is not only very well transferable to classical delivery logistics but also to dynamic technical service processes (Ulmer, 2017; Mardešić et al., 2023). For this thesis, the core message is therefore: ADP provides a scalable, anticipative, and practice-oriented decision framework for maintenance planning under uncertainty (Ulmer, 2019; Ulmer, 2020). In the context of charging infrastructure maintenance, this means that ADP can handle breakdowns that emerge during the day, prioritize interventions based on technical criticality, and allocate limited resources such as spare parts or technician time efficiently (Ulmer, Köster, & Mattfeld, 2018; Mardešić et al., 2023). The offline-online combination allows maintenance policies to be learned from historical failure data or simulations and then deployed in real-time dispatching systems (Ulmer et al., 2019).

# 4 Theoretical Framework

This chapter develops the formal framework used to represent the maintenance routing problem as a route-based Markov decision process (Ulmer et al., 2020), in which actions correspond to full route plans rather than individual node selections. Building on the operational setting introduced in Chapter 2, the objective is now to translate the planning problem into a mathematical structure that captures sequential decision-making under uncertainty. Particular emphasis is placed on the interaction between route construction, stochastic fault arrivals, time-dependent travel conditions, and limited daily service capacity. The chapter further explains why a state-action-value perspective is required and why approximate dynamic programming constitutes an appropriate solution methodology for the resulting problem. Table 5 in the Appendix summarizes the notation used throughout this chapter.

## 4.1 Markov Decision Process

The maintenance of electric vehicle charging infrastructure is inherently dynamic. At the start of the working day, not all information relevant for operational planning is yet known, because additional disturbances may occur during execution and traffic conditions may affect route feasibility. Consequently, the planning problem cannot be represented adequately by a purely static routing model. Instead, it is formulated as a Markov decision process (MDP), in which decisions are made sequentially and the system evolves stochastically over time.

A route-based MDP formulation is especially suitable in this context because route decisions have consequences beyond their immediate effect. Assigning a team to one area of the network influences the remaining time budget, the accessibility of future service jobs, and the ability to react to newly emerging disturbances. The purpose of the formulation is therefore not merely to choose the next stop, but to evaluate route decisions in light of their effect on the future development of the system.

### 4.1.1 Sets, indices, and temporal structure

The maintenance network consists of a set of charging stations $I = \{1, \ldots, 397\}$ and a set of service teams $K = \{1, 2\}$. The depot is indexed as node $0$, and the overall node set is given by $V = \{0\} \cup I$. Travel times between locations are denoted by $d_{ij}$ for all $i, j \in V$ and may vary by time of day. In addition, the charging stations are partitioned into zones $Z = \{1, \ldots, N_Z\}$, with $N_Z = 178$, which serve as an operational pre-structuring mechanism for route generation.

The decision process is embedded in a daily planning horizon. The working day begins at 8:00 and planning decisions are evaluated on an hourly grid $H = \{8, 9, \ldots, 16\}$. Within this structure, two types of decision points are distinguished. First, an initial planning decision is made at the beginning of the day, where a full route plan is constructed for both teams. Second, replanning decisions may occur during the day whenever new disturbances arise. This distinction between an initial plan and event-triggered route revisions reflects the operational logic of a maintenance system in which execution and planning must remain closely coupled.

Although each day constitutes a self-contained planning instance within this finite horizon, successive days are not independent. The days-since-maintenance variable $dsm_d^i$ and the carryover tasks from unresolved faults create an explicit inter-day coupling. Servicing a station on day $d$ lowers its fault probability on day $d + 1$, while leaving stations unattended raises the carryover backlog and increases expected future fault rates. The total planning objective therefore accumulates daily costs over a full operational year of $D$ days. Within a single day, the $Q^*$-function captures the downstream consequences of a decision for the remainder of that day; across days, these consequences propagate through the DSM update and the carryover mechanism.

The lunch break between 12:00 and 13:00 constitutes a deterministic component of the transition function rather than an explicit decision variable. It is inserted after the first stop whose scheduled departure falls at or after noon, shifting all subsequent arrival and departure times by one hour. Since the break applies uniformly and unconditionally, it requires no dedicated state variable or action representation, and is instead absorbed directly into the transition operator $P(s_{h+1} \mid s_h, a_h)$ as a fixed timing rule.

### 4.1.2 State representation

Let the system state at decision epoch $h \in H$ be denoted by

$$s_h = \big(\tau_h,\ X_h,\ Y_h,\ J_h,\ \Pi_h\big).$$

This representation is designed so that the state contains all information required to make the next decision. In that sense, it satisfies the Markov property: the probability distribution of the next state depends on the current state and the chosen action, but not on earlier system history beyond what is already encoded in $s_h$.
The first component, $\tau_h$, captures the current time within the workday and is measured in minutes since 8:00. The second component, $X_h$, describes the state of the service teams. For each team $k \in K$, this includes its current location $v_h^k$, its activity status $\sigma_h^k$, and the remaining time budget $b_h^k$ available before the route must return to the depot, defined as

$$b_h^k = T_{\max} - \tau_h - d_{v_h^k,0},$$

where $T_{\max}$ denotes the total length of the working day in minutes and $d_{v_h^k,0}$ is the travel time from the team's current location back to the depot. The activity status distinguishes whether a team is currently servicing a station, travelling, returning to the depot, or inactive. This part of the state is essential because route feasibility depends critically on the remaining usable time and on the team's current physical position.
The third component, $Y_h$, describes the station-related maintenance status. For each station $i \in I$, the state contains a maintenance indicator $m_h^i$, which records whether the required periodic maintenance has already been completed, and a variable $dsm_h^i$, representing the number of days since the last maintenance intervention. The latter variable plays a central role because it influences the probability of future disturbances. Importantly, the technical condition of a charging station is not represented here as a separate persistent binary or multi-level health state. Instead, disturbances are modeled as stochastic events that are immediately transformed into service requests.
The fourth component, $J_h$, captures the set of open maintenance jobs. It consists of three subsets: regular maintenance jobs $W_h^{\text{reg}}$, newly arrived fault jobs $W_h^{\text{fault}}$, and carryover fault jobs $W_h^{\text{carry}}$ transferred from the previous day. This distinction is operationally relevant because the planning problem must account simultaneously for preventive tasks, newly occurring corrective tasks, and accumulated backlog. The set of open regular maintenance jobs is directly determined by the maintenance indicators: $W_h^{\text{reg}} = \{i \in I : m_h^i = 0\}$.
Finally, $\Pi_h$ stores the planned routes of both service teams. For each team, the route plan contains a node sequence and the associated schedule of arrival and departure times. In a route-based formulation, including the route plan as part of the state is particularly useful because actions do not merely select an immediate successor node. Rather, they modify or reconstruct larger route segments and thereby influence the remaining structure of the entire working day.

### 4.1.3 Action space

The action space $A(s_h)$ depends on the type of decision point. At the start of the day, the decision consists of constructing an initial route plan for both teams. This process is divided into two stages. In the first stage, zones containing open stations are evaluated and filtered. The scoring can be based on classical geographic criteria such as average distance to the depot and zone area, or on a value-based priority measure reflecting maintenance urgency and station-specific relevance. Each team is then assigned a start zone, and stations are added to the candidate set by loading further zones in descending score order until a predefined maximum number of stations is reached.
In the second stage, the actual route plan is constructed. Based on the candidate sets and the carryover jobs inherited from the previous day, a feasible route for each team is generated by a greedy cheapest-insertion heuristic. Stations that cannot be accommodated within the remaining working time are dropped from the current day and remain open. This route construction procedure yields the full-day action taken at the initial decision point.
At replanning points during the day, the action space has a different interpretation. Newly arriving disturbance jobs are considered sequentially and inserted into the existing route of the team for which the additional insertion cost is smallest. The insertion cost accounts not only for the additional travel and labor expenditure but also for the downtime cost accumulated while the fault station waits for service. Since this downtime term grows with the waiting time until arrival, earlier insertion positions are almost always cheaper — meaning disruptions are in practice served as the next stop immediately after they are reported. If no feasible insertion exists because the remaining route time is insufficient, the disturbance is not ignored but transferred as a carryover job to the following day.
All actions must satisfy three feasibility conditions. First, each route must be temporally feasible, meaning that the planned return to the depot must occur within the allowable time horizon. Second, each team is required to end its route at the depot. Third, each station may be served by at most one team at any given time. These constraints ensure that every action corresponds to an operationally executable route plan. The chosen action at decision epoch $h$ is denoted $a_h \in A(s_h)$.

### 4.1.4 Transition dynamics

The transition function $P(s_{h+1} \mid s_h, a_h)$ describes how the system evolves between two consecutive decision points. During that interval, teams execute the planned route, travel to stations, perform maintenance or repair tasks, and update the completion status of serviced stations. Simultaneously, new disturbances may occur and travel times may deviate from their nominal values. As a result, the transition dynamics are stochastic even if the chosen action is fixed.
The route progression is determined by the planned arrival and departure schedule, but actual execution is affected by uncertain travel conditions. The formulation therefore allows for stochastic travel times, for example by using time-dependent lognormal travel-time realizations. This captures the operational reality that the duration of a trip depends not only on spatial distance but also on the time of day and traffic conditions.

Service completion updates the maintenance-related station state. A completed routine maintenance task sets the maintenance indicator to one and resets the days-since-maintenance variable to zero. A type-1 fault requires a standard repair with fixed duration $s_{\text{typ1}} = 60$ min and likewise resets the maintenance age. A type-2 fault entails a more extensive service duration that incorporates additional handling and depot-related effort through an aggregated time term. In the current route-based approximation, this depot-related part is not modeled as a separate physical route leg but is embedded directly in the service time. The aggregated service time for a type-2 fault at station $i$ is therefore given by

$$s_{\text{typ2}}(i) = s_{\text{dismount}} + d_{i,0} + s_{\text{handling}} + d_{0,i} + s_{\text{remount}},$$

with $s_{\text{dismount}} = 30$ min, $s_{\text{handling}} = 5$ min, and $s_{\text{remount}} = 30$ min. This modeling choice preserves the operational impact on route feasibility without introducing additional state and routing complexity.
At the end of each day $d$ — where $d$ indexes calendar days and $h \in H$ indexes decision epochs within a day — the days-since-maintenance variable is updated according to

$$dsm^i_{d+1} = \begin{cases} 0 & \text{if station } i \text{ received routine maintenance or fault repair on day } d, \\ dsm^i_d + 1 & \text{otherwise.} \end{cases}$$

Disturbance generation is modeled stochastically and depends on station-specific baseline probabilities, maintenance age, and station-level scaling factors. For each station $i \in I$ and fault type $\ell \in \{1, 2\}$, the per-hour fault probability at decision epoch $h$ is

$$p^i_\ell(h) = p_{\ell,\text{base}} \cdot f\left(dsm^i_h\right) \cdot \eta^i,$$

where $p_{\ell,\text{base}}$ is the baseline fault rate, $\eta^i$ is a station-level scaling factor, and $f(\cdot)$ is a recovery curve that captures how the fault probability increases with maintenance age:

$$f(t) = \lambda + (1 - \lambda) \cdot \frac{\min(t,\ \tau_{\text{rec}})}{\tau_{\text{rec}}}.$$

Here, $\lambda \in (0, 1)$ is the residual fault factor immediately after a maintenance visit and $\tau_{\text{rec}}$ is the number of days until full fault probability is restored. Faults that cannot be scheduled feasibly in the current day are carried over and become part of the next day's initial workload. Through this mechanism, unresolved work propagates over time and creates a dynamic coupling between successive planning periods.

### 4.1.5 Reward function

The optimization objective is formulated as a cost minimization problem. In MDP notation, the reward is therefore defined as the negative of the total cost incurred by an action and its resulting system evolution:

$$R = -\left(C_{\text{op}} + C_{\text{dt}}\right).$$

The total cost consists of an operational component and a downtime component:

$$C_{\text{op}} = n_{\text{active}} \cdot T_{\text{shift}} \cdot c_L + \sum_{(i,j)\,\in\,\text{trips}} \text{km}_{ij} \cdot c_F,$$

$$C_{\text{dt}} = \sum_{j\,\in\,\text{faults}} \Delta\tau_j^{\text{wait}} \cdot p_j \cdot c_{\text{dt}}.$$

The operational cost $C_{\text{op}}$ includes labor and travel expenditures. Labor costs depend on the number of active teams $n_{\text{active}}$ and the effective shift duration $T_{\text{shift}}$, while travel costs depend on the kilometers $\text{km}_{ij}$ traveled between successive route segments. In the present formulation, the labor cost parameter is $c_L = 35$ EUR per hour and the travel cost parameter is $c_F = 0.30$ EUR per kilometer.
The downtime-related cost $C_{\text{dt}}$ is accounted for on a per-day basis, summing over all unresolved fault jobs. Here $p_j$ denotes the nominal charging power of station $j$ in kilowatts and $c_{\text{dt}} = 0.50$ EUR per kilowatt-hour the downtime unit rate. The effective waiting time $\Delta\tau_j^{\text{wait}}$ depends on when the fault is resolved: for a fault that cannot be inserted into any team's route on the day it arrives, $\Delta\tau_j^{\text{wait}}$ equals the remaining workday duration; for a carryover fault resolved on the following day, it equals the elapsed workday time from the start of operations until service completion. A fault that remains unresolved over multiple days accumulates daily downtime costs on each affected day independently. This structure ensures that the objective does not reward short routes per se, but rather seeks an economically balanced compromise between efficient deployment and timely restoration of charging availability.
Overall, the reward function reflects the central trade-off of the maintenance problem. A route that minimizes travel distance may still perform poorly if it postpones high-impact repairs, while an aggressively reactive policy may generate excessive operating cost. The chosen reward structure therefore evaluates actions in terms of their total contribution to operational efficiency and service quality.
Together, the state space $S$, action space $A(s_h)$, transition function $P$, and reward $R$ constitute the route-based MDP. The planning objective is to find an optimal policy $\pi^* : S \to A$ that minimizes the expected total cost over the finite planning horizon. As formally defined in Section 4.2, this is expressed through the optimal state-action-value function $Q^*$:

$$\pi^*(s_h) = \arg\min_{a_h \in A(s_h)} Q^*(s_h, a_h).$$

## 4.2 State-Action-Value Structure

While the state representation describes the current condition of the system, it is not sufficient by itself to evaluate decision quality. In a dynamic maintenance routing problem, the attractiveness of a decision depends not only on its immediate cost, but also on how it shapes the future development of the system. This is the reason for adopting a state-action-value perspective.
The optimal state-action-value function $Q^*(s, a)$ assigns to each feasible action $a$ in state $s$

the minimum expected total downstream cost obtained when that action is chosen and the process then continues optimally. Formally, the optimal $Q$-function satisfies the Bellman equation

$$Q^*(s_h, a_h) = \mathbb{E}\left[ C(s_h, a_h) + \min_{a_{h+1} \in A(s_{h+1})} Q^*(s_{h+1},\, a_{h+1}) \,\middle|\, s_h,\, a_h \right],$$

where $C(s_h, a_h) = C_{\mathrm{op}} + C_{\mathrm{dt}}$ is the cost incurred in the current period and the expectation is taken over the stochastic transition $s_{h+1} \sim P(\cdot \mid s_h, a_h)$. The formulation is undiscounted ($\gamma = 1$), meaning that future costs are weighted equally to immediate costs — a deliberate choice reflecting that postponing a repair is equally costly regardless of when it occurs within the planning horizon. Conceptually, $Q^*$ combines two elements: the immediate cost of the action and the minimum expected future cost achievable from the resulting state. The latter component is crucial because route decisions alter the set of feasible responses to later disturbances.

This distinction is especially important in a route-based MDP. In a node-based formulation, an action might simply correspond to selecting the next customer or the next destination. In the present case, however, an action is more comprehensive: at the beginning of the day it defines a full route plan for both teams, and during the day it may modify substantial parts of the remaining route through a disturbance insertion. Consequently, the quality of an action cannot be assessed only by looking at its immediate incremental cost. It must instead be judged in terms of its impact on the remainder of the route and on the system's future flexibility.

The practical meaning of this is straightforward. Inserting a station into the route consumes time, changes the expected depot return time, affects accessibility of later jobs, and may either preserve or reduce the ability to react to future disturbances. A locally attractive decision may therefore be globally poor if it leaves too little slack for high-priority faults that are likely to appear later. Conversely, a locally more expensive insertion can be preferable when it maintains route adaptability and reduces expected future service delay.

The $Q$-structure provides the conceptual foundation for evaluating such trade-offs. It translates the routing problem from a purely myopic assignment logic into a sequential decision problem in which future consequences are explicitly recognized. In this way, the state-action-value perspective forms the bridge between the route-based MDP formulation and the approximation strategy introduced in the next section.

## 4.3 ADP Foundations and the Offline-Online Approach

Although the route-based MDP provides an appropriate conceptual representation of the maintenance problem, solving it exactly is computationally infeasible. The dimensionality of the state space is very large because it combines temporal information, two vehicle states, station-specific maintenance information, multiple categories of open jobs, and the current route plan. The action space is equally large, since route construction and route modification decisions can take many feasible forms. In addition, the transition process is stochastic due to fault arrivals and uncertain travel times.

For these reasons, the problem is addressed through approximate dynamic programming (ADP). The central idea of ADP is not to compute the exact optimal value function for every possible state, which would be impractical, but to approximate the value of future decisions in a computationally tractable way. This approximation can be based on simulation, parameterized value estimates, structured heuristics, or combinations thereof. The purpose is to obtain a policy that is sufficiently accurate to support high-quality operational decisions while remaining fast enough for day-to-day deployment.

In the present context, ADP is particularly attractive because the decision problem unfolds in two distinct modes: planning at the beginning of the day and replanning during execution. A suitable framework must therefore support both anticipatory route construction and rapid reaction to new disturbances. This naturally leads to an offline-online structure.

In the offline phase, the decision logic is prepared before actual daily operation. This can include calibration of scoring parameters, design of value surrogates, generation of representative simulation scenarios, and estimation of how route characteristics translate into future cost consequences. The offline phase is therefore the stage at which structural knowledge about the maintenance system is embedded into the approximation framework.

In the online phase, the system observes the current state and must select a feasible action under tight time constraints. The precomputed or learned information from the offline stage is then used to guide route construction or route revision. The degree of online computation varies across policies: simpler approaches rely entirely on precomputed parameters for fast online decisions, while more advanced approaches additionally perform targeted online simulation — using the precomputed base policy — to refine those decisions. In both cases, the offline preparation is what makes the approach practical for operational use.

The route-based logic of the present model fits this ADP structure particularly well. The zonal pre-selection, candidate generation, and greedy insertion rules should not be interpreted merely as ad hoc heuristics. Within the ADP framework, they can be viewed as operational policy components that exploit structural information about the value of different routing choices. Their role is to restrict the search space intelligently and to approximate the effect of decisions on future system performance.

Taken together, ADP provides the methodological layer that makes the route-based MDP operationally usable. The route-based MDP defines the decision problem rigorously, the state-action-value structure clarifies what has to be approximated, and the offline-online ADP scheme supplies a practical mechanism for producing fast and robust route decisions under uncertainty. Chapter 5 translates this framework into concrete policies: ranging from a myopic baseline that ignores future costs entirely, through cost function approximations that embed learned station values into the routing score, to a rollout-based lookahead that approximates the value function via short-horizon Monte Carlo simulation.

# 5 Solution Methodology

This chapter presents the solution methodology developed for the maintenance planning problem of electric charging stations. The methodological framework is designed to handle a dynamic and stochastic environment in which new failures may occur during the day and existing routes must be adjusted accordingly. Since the problem is too complex to solve exactly at the operational level, the thesis relies on a hierarchy of heuristic policies and approximate dynamic programming-based approaches.

The methodology starts with a common spatial decision layer that determines which zones of the network are considered first. On top of this shared structure, different decision policies are applied. The simplest policy is the Myopic baseline, which relies only on the currently available information. Myopic+ extends this baseline by incorporating a stronger economic prioritization. Cost Function Approximation introduces a learned station value that captures the future cost of postponing service. Dynamic Balance extends this value-based structure by introducing a state-dependent routing parameter. Value Function Approximation extends this hierarchy by adding an online rollout layer that replaces the greedy drop decision with a simulation-based lookahead.

The purpose of this chapter is to describe these policies in a consistent way and to make clear how they differ in terms of decision logic, prioritization, and replanning behavior. This structure also allows for a fair comparison of the methods in the computational experiments.

## 5.1 Shared Zone Selection Logic

The zone selection mechanism forms a preliminary decision layer that acts before the actual routing and disturbance handling logic. Its purpose is to structure the spatial search space and to determine which parts of the service area are addressed first on a given day. All policies in this thesis share the same underlying zone structure, based on a K-Means decomposition of the station network, but differ in how zones are prioritized.

Zone selection is evaluated as an independent experimental dimension: each of the four base policies is tested under both a centrality-based and a value-based mode. This design allows the effect of the zone selection criterion to be isolated from the effect of the routing and replanning policy.

In the centrality-based mode, zones are scored according to their geographic position within the network. Specifically, the score of a zone is derived from its mean distance to the centroids of all other zones:

$$\text{score}(z) = 1 - \frac{\bar{d}_z - \min_z \bar{d}_z}{\max_z \bar{d}_z - \min_z \bar{d}_z},$$

where $\bar{d}_z$ denotes the mean distance from zone $z$'s centroid to the centroids of all other zones. Zones that are geographically central within the network, that is, close to many other zones, receive a high score and are therefore processed first. This criterion is policy-agnostic and serves as the baseline for zone prioritization.

In the value-based mode, the zone score additionally incorporates the expected maintenance relevance of the stations it contains. Specifically, the score combines a policy-specific station value, aggregated over all open stations in the zone, with the depot distance term.

The station value function differs by policy: Myopic and Myopic+ use a power-weighted urgency measure based on days since maintenance, while CFA and Dynamic Balance use the learned cost approximation $\tilde{C}$. This allows the zone selection to reflect not only geographic structure but also the anticipated cost consequences of postponing service at the stations within a zone.

## 5.2 Myopic

The Myopic policy serves as the baseline heuristic for all subsequent methods. It makes all decisions solely on the basis of currently available information and does not incorporate any form of lookahead or learned cost approximation. This makes it a fast, transparent, and easily interpretable reference point for evaluating the added value of the more advanced policies.

#### Initial Plan

At the beginning of the day, after zone selection and candidate set generation, the initial route is constructed by a greedy routing procedure. Starting from the depot, the next station is always chosen as the one with the highest score among all remaining candidates, where the score is defined as the inverse of the travel time from the current position:

$$\text{score}(i, \mathrm{cur}) = \frac{1}{\max(1,\, d_{\mathrm{cur},i})}.$$

This rule prioritizes nearby stations and produces a compact, travel-time-efficient route without any economic weighting. Carryover tasks from previous days are always inserted first, using a nearest-neighbor rule, before any routine tasks are scheduled. This ensures that previously unresolved faults are addressed as early as possible in the new day. Routine stations that cannot be accommodated within the remaining workday are dropped and remain open for the following day.

#### Disruption Handling

When new disruptions arrive during the day, each disruption is evaluated for insertion into the current route using a greedy cheapest-insertion procedure. For every combination of team and insertion position, the additional cost is computed as:

$$c_{\mathrm{ins}} = \frac{\Delta t_i + s_i}{60} \cdot c_L + \max(0,\, \Delta \ell_i) \cdot c_F + \frac{\max(0,\, t_i^{\mathrm{arr}} + s_i - r_i)}{60} \cdot p_i \cdot c_{\mathrm{dt}},$$

where the detour time and detour distance are defined as

$$\Delta t_i = d_{\mathrm{prev},i} + d_{i,\mathrm{next}} - d_{\mathrm{prev},\mathrm{next}}, \qquad \Delta \ell_i = \ell_{\mathrm{prev},i} + \ell_{i,\mathrm{next}} - \ell_{\mathrm{prev},\mathrm{next}},$$

$s_i$ is the service time of the disruption in minutes, $c_L = 35$ EUR/h the labor rate, $c_F = 0.30$ EUR/km the fuel rate, $t_i^{\mathrm{arr}}$ the arrival time at the disrupted station in minutes since

8:00, $r_i$ the fault report time in minutes since 8:00, $p_i$ the rated power in kW, and $c_{\mathrm{dt}} = 0.50$ EUR/kWh the downtime cost rate. Here $d_{ij}$ denotes travel time in minutes between nodes $i$ and $j$ as defined in Chapter 4, and $\ell_{ij}$ the corresponding travel distance in kilometers. Disruptions are processed in ascending order of their cheapest insertion cost, so the least expensive disruption is inserted first.

If no feasible direct insertion exists, a drop-and-insert fallback is applied. All remaining routine stops are ranked by their detour time $\Delta t_i$, defined as the travel time saved by removing stop $i$ from the current route. Stops are dropped in ascending order of the score $1/\max(1, \Delta t_i)$, meaning the stop that saves the largest detour is removed first. Drops are applied iteratively until the disruption can be feasibly inserted. Dropped stops are carried over to the following day. If no combination of drops creates sufficient capacity, the disruption itself becomes a carryover.

## 5.3 Myopic+

Myopic+ shares the same greedy architecture as Myopic but replaces the purely travel-time-based scoring with a power-weighted score. The central idea is that stations with higher rated power should be visited earlier in the day, since a failure at such a station incurs higher downtime costs. This economic weighting is applied consistently in both the initial plan and the drop decision during disruption handling.

### Initial Plan

Carryover tasks are inserted first using nearest-neighbor, identical to Myopic. For routine tasks, the next station is selected by the highest score among remaining candidates, where the score now weights travel time by rated power:

$$\text{score}(i,\, \mathrm{cur}) = \frac{p_i}{\max(1,\, d_{\mathrm{cur},i})}.$$

A station with high rated power $p_i$ and short travel time $d_{\mathrm{cur},i}$ receives the highest priority. Compared to Myopic, a more distant high-power station can therefore be preferred over a nearby low-power station if the power advantage outweighs the travel penalty.

### Disruption Handling

The insertion cost formula for evaluating disruption candidates is identical to Myopic — the same full economic cost over detour time, distance, and downtime is computed for every (team, position) pair, and disruptions are processed in ascending order of insertion cost.

The key difference lies in the drop score. When no direct insertion is feasible, remaining routine stops are ranked by:

$$\text{drop\_score}(i) = \frac{p_i}{\max(1,\, \Delta t_i)},$$

where $\Delta t_i$ is the detour time saved by removing stop $i$. Stops are dropped in ascending order of this score — low-power stations with large detours are removed first, while high-power stations or those with small detours are protected. This ensures that economically important stations remain in the route whenever capacity is tight. Dropped stops are carried over to the following day; if no combination of drops resolves the capacity shortage, the disruption itself becomes a carryover.

## 5.4 Cost Function Approximation

CFA extends the greedy decision logic by introducing a learned approximation of the future cost of postponing a station. Instead of scoring stations purely by proximity or rated power, CFA estimates how costly it would be to leave a station unserved and uses this estimate as a prioritization signal throughout the planning day. The approximation is linear in a small set of station-specific features and trained offline, while all operational decisions remain fully greedy.

### Cost Approximation and Training

CFA represents the cost of leaving station $i$ unserved through a linear value function over four features:

$$\phi(i) = \big[p_i,\ a_i,\ \rho(\mathrm{dsm}_i),\ \bar{d}_i\big]^\top,$$

where $p_i$ is the rated power, $a_i$ the station age in years, $\rho(\mathrm{dsm}_i)$ the recovery curve evaluated at the days since last maintenance, and $\bar{d}_i$ the mean distance from station $i$ to all other stations, precomputed statically at initialization. The recovery curve captures the temporal failure risk: it is lowest immediately after maintenance and rises back toward its maximum as time passes. Before entering the model, all features are standardized with training means and standard deviations, and the value function is:

$$\tilde{C}(i,\, \mathrm{dsm}_i) = \theta^\top \phi_{\mathsf{scaled}}(i,\, \mathrm{dsm}_i).$$

The weight vector $\theta \in \mathbb{R}^4$ is estimated offline via Ridge regression on contrastive training labels. Each label is the discounted cost difference between two simulated suffix trajectories for station $i$: one in which the station is served and one in which it is dropped. This makes $\tilde{C}$ a direct approximation of the marginal value of servicing a station rather than a fit to aggregate rollout costs. The standardization parameters $\mu$ and $\sigma$ are stored together with $\theta$ and loaded at inference time.

### Initial Plan

Carryover tasks are inserted first using nearest-neighbor, as in Myopic and Myopic+. For routine tasks, the station selection score combines the learned value with travel time:

$$\mathsf{score}(i,\, \mathrm{dsm}_i,\, \mathrm{cur}) = \frac{\tilde{C}(i,\, \mathrm{dsm}_i) + s}{\max(1,\ d_{\mathrm{cur},i})},$$

where $s = \max(0, -\min_i \tilde{C}) + 1$ is a shift that ensures all scores are strictly positive, preventing negative $\tilde{C}$ values from reversing the travel-time penalty. When value-based zone selection is active, CFA uses $\tilde{C}$ as the station value function passed to the zone selector, as described in Section 5.1.

### Disruption Handling

The insertion cost formula is identical to Myopic — full economic costs over detour time, distance, and downtime are evaluated for each (team, position) pair, and disruptions are processed in ascending order of insertion cost. When no direct insertion is feasible, remaining routine stops are ranked by a drop score that trades off future value against the detour savings from removal:

$$\text{drop\_score}(i, \mathrm{dsm}_i) = \tilde{C}(i, \mathrm{dsm}_i) - \tfrac{c_L}{60} \cdot \Delta t_i,$$

where $c_L/60 = 35/60$ EUR/min is the per-minute labor rate and $\Delta t_i$ is the detour time saved by removing stop $i$. Stops are dropped in ascending order — stations with low future value and large detour savings are removed first. Dropped stops become carryovers; if no combination of drops creates sufficient capacity, the disruption itself is carried over to the following day.

## 5.5 Dynamic Balance

Dynamic Balance builds directly on CFA and extends it by introducing a state-dependent balancing parameter $\delta$. The purpose of this policy is to dynamically trade off the economic importance of a station, captured by the approximated value $\tilde{C}(i)$, against routing efficiency, captured by the distance term in the greedy selection rules. In contrast to CFA, where this trade-off is fixed, Dynamic Balance allows the emphasis to shift with the state of the system.

### Balance Concept and Parameter

Following the Dynamic Balance concept of Stein, routing quality should not be determined by a single static priority rule. Instead, the decision logic should balance two competing objectives: the servicing of economically important stations and the maintenance of spatial efficiency in the route. This trade-off is particularly relevant in a dynamic maintenance setting, where the best choice early in the planning horizon may not be the best choice as the horizon closes.

The balance parameter $\delta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ controls the relative strength of the distance term in the greedy score. The special case $\delta = 0.5$ yields a routing score with a linear distance penalty, which is identical to CFA. Smaller values of $\delta$ reduce the importance of distance and shift the focus more strongly toward the approximated value $\tilde{C}(i)$. Larger values increase the distance penalty and thereby place greater emphasis on routing compactness. The parameter is determined once before the start of each day and remains constant throughout the day. In the rule-based mode used in the current setup, $\delta$ follows a simple binary threshold: it is set to $\delta = 0.5$ — identical to CFA — as long as at least 130 stations remain unserved, and switches to $\delta = 0.9$ once fewer than 130 stations remain. In practice, this means the policy behaves like CFA throughout most of the planning horizon and shifts toward more compact routing only in the final stage.

#### Initial Plan

Dynamic Balance inherits all core components from CFA without modification: the feature vector $\phi(i)$, the learned value function $\tilde{C}(i, \mathrm{dsm}_i)$, the shift $s$, the carryover nearest-neighbor insertion, and the value-based zone selection. The only change is in the routing score, where the distance exponent is now controlled by $\delta$:

$$\text{score}(i,\, \mathrm{dsm}_i,\, \mathrm{cur}) = \frac{\tilde{C}(i,\, \mathrm{dsm}_i) + s}{\max(1,\ d_{\mathrm{cur},i})^{2\delta}}.$$

When $\delta = 0.5$, the exponent equals one and the formula reduces exactly to the CFA score. For $\delta < 0.5$, distance is penalized less than linearly, so high-value stations can be reached at a greater travel cost. For $\delta > 0.5$, the distance penalty grows super-linearly, favoring compact routing over value-based prioritization.

#### Disruption Handling

The insertion cost formula is identical to CFA and Myopic. When no direct insertion is feasible, routine stops are removed according to a drop score that mirrors the same $\delta$-controlled balance:

$$\text{drop\_score}(i,\, \mathrm{dsm}_i) = \tilde{C}(i,\, \mathrm{dsm}_i) - 2\delta \cdot \tfrac{c_L}{60} \cdot \Delta t_i,$$

where $\Delta t_i$ is the detour time saved by removing stop $i$ and $c_L/60 = 35/60$ EUR/min. The factor $2\delta$ scales the weight of the detour savings relative to the future value. At $\delta = 0.5$ this reduces to the CFA drop score. For larger $\delta$, stops with high detour savings are dropped more aggressively, improving feasibility at the cost of dropping potentially valuable stations.

## 5.6 Value Function Approximation

Value Function Approximation is not an independent planning algorithm in this thesis, but an online rollout layer that is placed on top of an existing base heuristic. The base policy remains fully intact and continues to generate the operational plan. VFA only intervenes at selected decision points, where a rollout simulation indicates that an alternative action yields a better expected future outcome. In this sense, VFA is an offline-online hybrid strategy in which the heuristic structure is defined in advance and the improvement is obtained online through simulation-based evaluation.

### Offline-Online Architecture

The VFA framework is based on a strict separation between a static offline component and a dynamic online decision component. In the offline part, the chosen base policy is fixed and prepared for use in the rollout layer. In the online part, the current decision state is evaluated by simulating alternative actions and comparing their expected future consequences over a finite horizon. The central idea is not to replace the base policy, but to complement it with a future-oriented evaluation mechanism. This separation allows the method to combine computational tractability with a stronger anticipation of future events. The offline-online structure is important because it keeps the operational logic simple while still enabling more informed decisions at critical points. The offline policy provides the reference behavior, whereas the online rollout approximates the value of different actions by simulating their downstream effects. This is consistent with the general ADP principle that a good heuristic base policy can be improved by a local lookahead layer. In the context of this thesis, this architecture makes it possible to retain the strengths of the previously defined policies while adding a more anticipatory decision mechanism.

### Offline Phase: Base Policy

In the offline phase, a base policy is selected as the foundation for the rollout layer. Depending on the experimental setup, this base policy may be Myopic, Myopic+, CFA, or Dynamic Balance. The chosen policy remains unchanged during the rollout process and continues to handle the standard operational planning. The role of the offline phase is therefore to provide a stable and fully specified heuristic structure on which the rollout can build.
The base policy is accessed through a unified interface, which allows the VFA layer to remain independent of the concrete heuristic used underneath. This is important because it makes the rollout mechanism modular and transferable across different policy classes. For the initial plan, the base policy remains fully responsible in the standard configuration. In the standard setting, the offline phase does not perform any additional optimization itself; rather, it defines the behavior that will later be evaluated and, if necessary, overridden by the online layer.
This means that VFA does not construct routes from scratch. Instead, it uses the base policy as the default operational rule and improves it only when the rollout suggests a better local alternative. The offline phase is therefore best understood as the structural backbone of the method. It guarantees that the system always has a feasible and interpretable default behavior, even if the rollout layer is not activated or cannot find a superior candidate.

### Online Phase: Rollout

The actual value approximation takes place in the online phase through rollout simulations. VFA is activated when a newly reported disturbance cannot be inserted directly and a drop decision must be made. The base policy ranks all remaining routine stops by its drop score and identifies the top-$k$ candidates. For each candidate, the system constructs a post-decision state — the system state after that drop has been applied but before future

uncertainty materializes — and simulates the remaining planning horizon using the base policy.
To ensure a fair comparison across candidates, all candidates are evaluated under the same set of stochastic scenarios, generated using common random numbers. This reduces sampling noise and makes the cost estimates statistically comparable. The candidate with the lowest expected horizon cost is selected and applied, overriding the base-policy choice whenever a better alternative is found. If the rollout computation exceeds a predefined time budget, the system falls back to the greedy base-policy decision.
Overall, VFA represents the highest level in the methodological hierarchy of this thesis. It combines the robustness of a heuristic base policy with the anticipatory power of simulation-based lookahead. Compared with CFA and Dynamic Balance, it is more strongly future-oriented because it does not merely approximate station relevance, but directly evaluates the expected downstream cost of alternative drop decisions. At the same time, it remains operationally feasible because rollout is applied only at drop decision points rather than throughout the entire planning process.

# 6 Experimental Setup

This chapter specifies the simulation environment, data basis, and evaluation metrics used to compare the maintenance policies. The setup is designed to capture both the operational structure of the charging-station maintenance problem and the stochastic effects that influence routing, service times, and downtime.

## 6.1 Simulation Design

The policy evaluation is based on an event-driven day simulation that represents the daily operations of a maintenance service for electric charging stations in Würzburg over a horizon of up to 365 days. The simulation is repeated in Monte Carlo experiments so that random disruptions and travel times can be assessed statistically rather than through a single deterministic run. This is necessary because both failure generation and travel times contain stochastic elements.
Each simulation day begins at 8:00 a.m. with route initialization for two maintenance teams. The daily planning process first assigns starting zones and incorporates unresolved tasks carried over from previous days, after which a policy-specific initial route is constructed. During the working period, new disruptions may occur, the teams advance along their routes, and the current plan may be updated if the selected policy triggers a re-optimization step. The day ends with the calculation and logging of operational and downtime-related costs.
For robustness, each policy is evaluated across multiple independent runs with separate random seeds. The random streams for disruption generation and travel times are kept separate to avoid unintended dependence between these uncertainty sources. After all runs have been completed, the results are aggregated by policy using descriptive statistics

such as mean, standard deviation, minimum, and maximum.

## 6.2 Routing and Travel Times

Travel times between the depot and the 397 charging stations are derived from Open Source Routing Machine data based on OpenStreetMap extracts for Bavaria. A complete 398 by 398 travel-time matrix is precomputed for the depot and all stations and then reused during simulation to avoid repeated routing requests. In addition, hourly traffic matrices are generated to reflect time-dependent congestion effects during the workday from 8:00 to 17:00. To represent short-term variability in traffic conditions, the deterministic travel times are further perturbed by a log-normal stochastic component. This makes it possible to evaluate whether a route remains feasible under realistic travel-time fluctuations rather than only under average conditions. The resulting route robustness is summarized through indicators such as mean return time, 95th-percentile return time, overtime probability, and expected overtime minutes.

## 6.3 Data Basis

The simulation is based on a real data set of publicly available charging infrastructure in the city of Würzburg, obtained from the municipal open data portal (Stadt Würzburg, 2024). The portal provides station-level records for the entire public charging network in Würzburg, including geographic coordinates, charging power, number of charging points, and commissioning date. After filtering for operational stations, the data set contains 397 charging stations, consisting of both normal charging stations and fast-charging stations. Station age is inferred from the commissioning date relative to the simulation reference date.

The depot is located at the WVV maintenance facility in Sanderau, which serves as the operational base for both maintenance teams. To support the daily planning process, the city area is additionally structured into 178 spatial zones using clustering-based preprocessing. These zones provide a geographical abstraction that helps select promising daily work areas and organize station visits efficiently.

Disruptions are modeled stochastically on a station-by-station and hour-by-hour basis. The probability of failure increases with time since the last maintenance intervention and is further modified by station-specific characteristics such as charger type and age. In this way, the simulation reflects both operational deterioration and heterogeneity across the charging network.

## 6.4 Evaluation Metrics

The evaluation focuses on four groups of metrics: cost, service performance, backlog accumulation, and route robustness. Cost metrics combine labor costs, fuel costs, and downtime costs, allowing the total economic burden of each policy to be measured in euros. Labor costs are derived from working time, fuel costs from driven distance, and downtime costs from the monetary valuation of unavailable charging capacity.

Service performance is measured through indicators that capture how quickly the network is covered and how effectively disruptions are handled. Relevant measures include the number of days until all stations have been serviced at least once, the total number of generated disruptions, and the share of disruptions resolved on the same day. The same-day resolution rate is particularly important because it reflects the responsiveness of the maintenance system under stochastic demand.

Backlog behavior is captured through carryover metrics. These metrics record how many unresolved disruptions remain at the end of a day and how strongly such tasks accumulate over time. A sustained increase in carryover would indicate that a policy is no longer able to keep pace with incoming disruptions.

Route robustness is assessed through the stochastic travel-time indicators introduced in Section 6.2: mean return time, 95th-percentile return time, overtime probability, and expected overtime minutes. These measures quantify how reliably a planned route can be completed within the available workday under realistic travel-time variability.

Taken together, these metrics provide a structured basis for comparing the considered policies under identical stochastic conditions. They allow the analysis to cover not only total cost, but also operational feasibility, responsiveness, system stability, and schedule robustness.

# 7 Results and Analysis

This chapter evaluates the performance of the four policies developed in Chapter 5 across two experimental dimensions. Section 7.1 compares the four policy mechanisms under a controlled zone-selection setting to isolate the effect of the routing and prioritization logic. Section 7.2 then examines the role of zone selection itself, which turns out to be the dominant cost driver. Section 7.3 evaluates the Rolling Horizon rollout layer, and Section 7.4 provides statistical validation of all key comparisons. All results are based on 500 Monte Carlo runs per variant using seeds 1–500. Because each seed generates an identical stochastic disturbance sequence across all variants, pairwise comparisons are matched by seed, which substantially reduces noise.

## 7.1 Policy Mechanism Comparison

### Setup

Four policies are compared: Myopic (greedy cheapest insertion, no value weighting), Myopic+ (routine tasks and drop decisions scored by station power), CFA (cost function approximation with learned parameter vector $\theta$, scoring by $\tilde{C}/$distance), and DB (CFA extended by a state-dependent balance parameter $\delta$ that modulates the distance exponent). To isolate the effect of the routing and prioritization logic from zone-selection effects, all four policies are evaluated under the same centrality-based zone selection in this section. Centrality assigns zones by geographic proximity to the depot and convex hull area, without any policy-specific value signal.

**Results**

Table 1 summarizes the results. Under centrality-based zone selection, all four policies perform nearly identically. Mean total costs range from € 40,183 (DB) to € 40,410 (Myopic) – a spread of just € 227 or 0.6 %, which is well within stochastic noise. Same-day rates differ by less than 0.3 percentage points across all policies (67.3 %–67.6 %), and mean carryover tasks per simulation are similarly close (52.8–53.4). The entire cost difference is driven by downtime costs, yet even here the range is narrow (€ 16,642–€ 17,141). Labor and travel costs are virtually identical across all four policies.

Table 1: Policy comparison under centrality-based zone selection (500 runs each).

| Policy | Mean Total Cost (€) | SD (€) | Mean Downtime Cost (€) | Same-Day Rate |
|---|---|---|---|---|
| Myopic | 40,410 | 5,134 | 17,141 | 67.3 % |
| Myopic+ | 40,144 | 5,197 | 16,642 | 67.5 % |
| CFA | 40,187 | 5,156 | 16,695 | 67.6 % |
| DB | 40,183 | 5,128 | 16,702 | 67.4 % |

Figure 4 visualizes the cost distributions. The four policies show almost completely overlapping distributions; no consistent separation is visible. Figure 5 plots the same-day rate against total cost per run. All four policies occupy essentially the same region, with no policy forming a distinct cluster.

**Discussion**

The near-identical performance of all four policies under centrality is the central finding of this section. All four share the same greedy cheapest-insertion backbone and differ in two respects: how routine tasks are scored in the initial daily plan (the `route_score_fn`), and how drop candidates are ranked during replanning (the `drop_score_fn`). Under centrality-based zone selection, both teams are assigned the same type of zone regardless of policy, so the station pool available to each team is essentially identical. In this setting, a more sophisticated value-weighted scoring function yields essentially no benefit: the zone already determines which stations are available, and the different orderings produced by power-weighted versus distance-only scoring have little aggregate effect. The decisive question is therefore not *how* a policy routes within a zone, but *which* zone it selects.

Even Myopic, which assigns no priority to high-power stations during replanning, performs within 0.6 % of the advanced policies. While its lack of a prioritization mechanism causes slightly higher downtime costs on the most economically valuable stations, this effect is

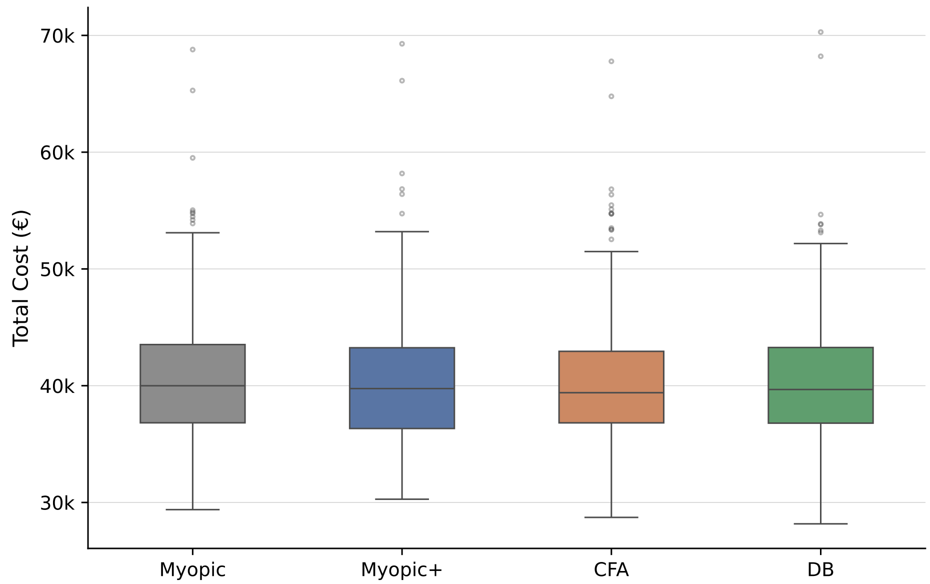


Figure 4: Total cost distributions for all four policies under centrality-based zone selection (500 runs each). Boxes show interquartile range; whiskers extend to 1.5×IQR.

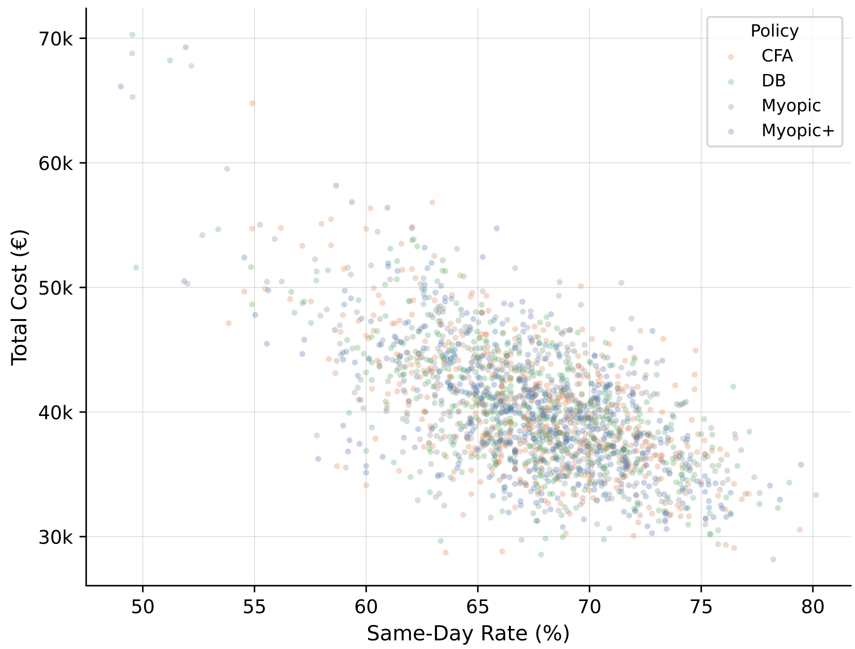


Figure 5: Same-day rate versus total cost for all individual runs under centrality-based zone selection. Each point represents one Monte Carlo run; color indicates policy.

small when the policy has no influence over which zones are visited at all. This observation motivates the analysis in Section 7.2.

## 7.2 Impact of Zone Selection on Cost

### Setup

Section 7.1 showed that all four policies perform nearly identically under centrality-based zone selection. This section asks whether the choice of zone-selection strategy itself matters – and if so, by how much. Both zone-selection modes are evaluated for all four policies: centrality (geographic proximity to depot and convex hull area) and value-based (zones ranked by expected economic impact, computed via the policy's own value signal). All other simulation parameters are held constant.

### Results

Table 2 reports mean total costs and cost components for all eight variants. Switching from centrality to value-based zone selection reduces mean total costs substantially across all four policies: by € 5,525 (13.7 %) for Myopic, € 5,831 (14.5 %) for Myopic+, € 6,394 (15.9 %) for CFA, and € 6,512 (16.2 %) for DB. In absolute terms, all four policies converge on costs of € 33,671–€ 34,885 under value-based selection, compared to € 40,144–€ 40,410 under centrality.

Table 2: Cost comparison across zone-selection modes for all four policies (500 runs each). $\Delta$ is computed relative to centrality for each policy.

| Policy | Zone Selection | Mean Total (€) | SD (€) | Mean Downtime (€) | Mean Labor (€) | $\Delta$ (%) |
|---|---|---|---|---|---|---|
| Myopic | centrality | 40,410 | 5,134 | 17,141 | 22,098 | – |
| Myopic | value-based | 34,885 | 4,824 | 12,114 | 21,577 | −13.7 % |
| Myopic+ | centrality | 40,144 | 5,197 | 16,642 | 22,286 | – |
| Myopic+ | value-based | 34,313 | 4,727 | 11,556 | 21,546 | −14.5 % |
| CFA | centrality | 40,187 | 5,156 | 16,695 | 22,271 | – |
| CFA | value-based | 33,793 | 4,525 | 11,111 | 21,520 | −15.9 % |
| DB | centrality | 40,183 | 5,128 | 16,702 | 22,267 | – |
| DB | value-based | 33,671 | 4,546 | 11,022 | 21,491 | −16.2 % |

Figure 6 shows the full cost distributions. The downward shift from centrality to value-based is consistent across all 500 runs and all four policies, confirming that the effect is not driven by a small subset of favorable scenarios. Figure 7 decomposes mean costs into labor, travel, and downtime components. Labor and travel costs are virtually unchanged across all eight variants (approximately € 22,000 and € 1,200 respectively). The entire reduction comes from downtime costs, which fall by roughly 29–34 % across all policies.

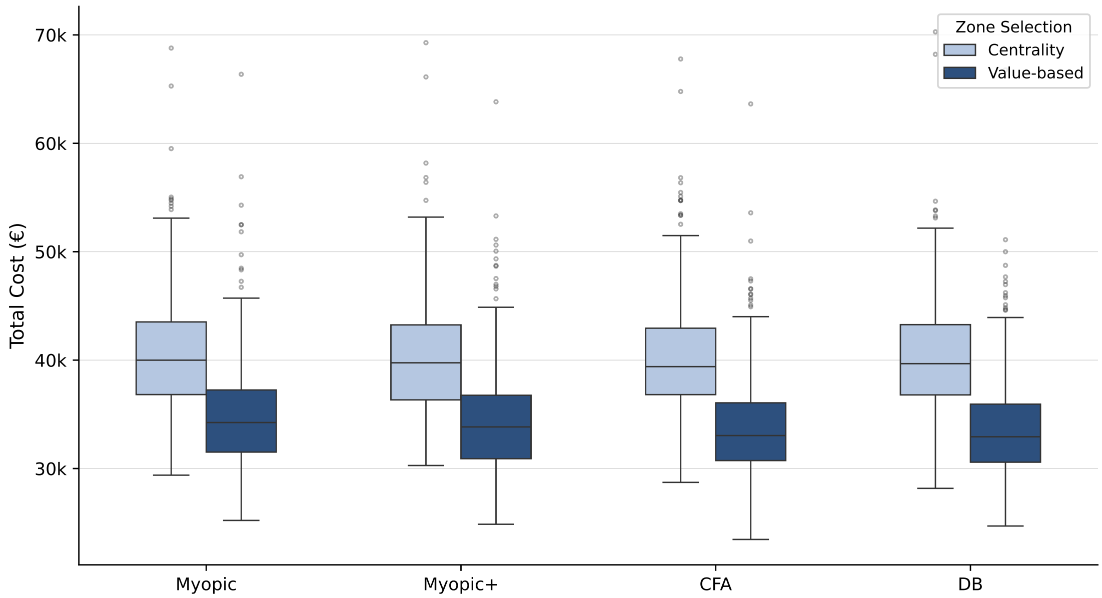


Figure 6: Total cost distributions for all four policies under centrality (light) and value-based (dark) zone selection (500 runs each). Boxes show interquartile range; whiskers extend to 1.5×IQR.

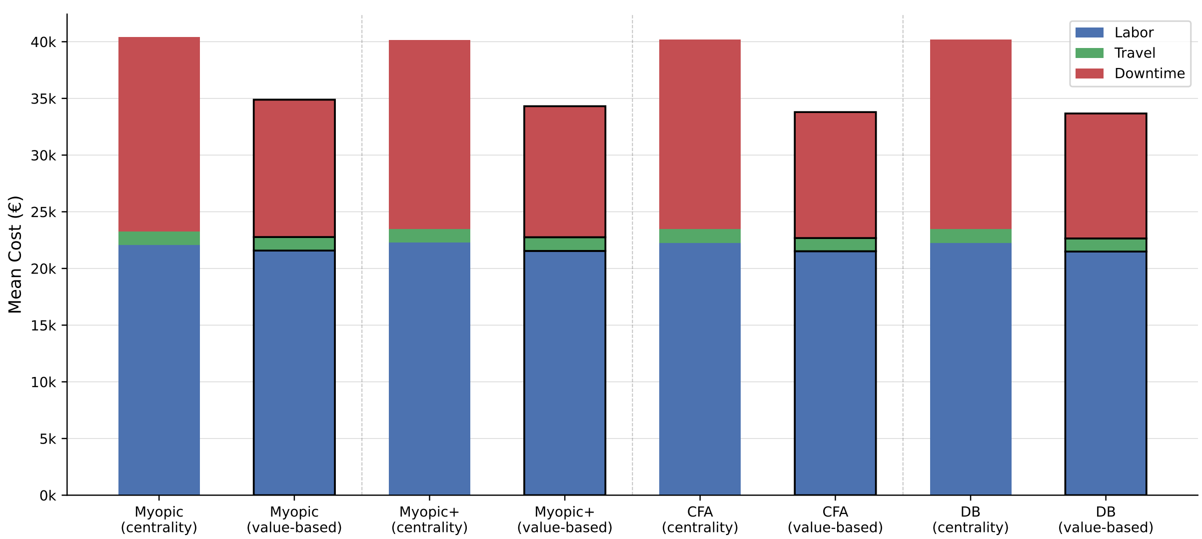


Figure 7: Mean cost components (labor, travel, downtime) per policy and zone-selection mode. Bars with black border correspond to value-based selection. Dashed vertical lines separate policy groups.

### Discussion

The mechanism behind the zone-selection effect is straightforward: value-based selection prioritizes zones containing stations with high rated power and long elapsed time since last maintenance, which are precisely the stations generating the highest downtime costs per hour of unavailability. By visiting these zones earlier in the simulation, the policies resolve high-cost faults before they accumulate multiple days of downtime charges. The routing effort – labor and travel – is unaffected because the total number of stations visited per day remains the same.

A striking aspect of the results is that Myopic captures most of the zone-selection gain despite having no learned value function: it saves 13.7 % versus 16.2 % for DB. This confirms that the dominant effect operates at the zone-selection level, not within the routing logic. Even a policy that is agnostic to station value at the routing stage captures most of the available gain simply by being directed toward the right zones. A residual gap between Myopic and the advanced policies remains under value-based selection (€ 34,885 versus € 33,671–€ 34,313); statistical analysis in Section 7.4 shows this gap is detectable but small.

Compared to the policy differences observed in Section 7.1 (less than 1 % among all four policies under centrality), the zone-selection effect of 14–16 % is dramatically larger. Zone selection completely dominates the policy mechanism as a cost driver in this setting. Section 7.3 examines whether the Rolling Horizon rollout layer can provide additional gains on top of value-based zone selection.

## 7.3 VFA Rollout: Performance and Limitations

### Setup

Sections 7.1 and 7.2 established that value-based zone selection is the dominant cost driver, and that all four policies perform nearly identically under centrality-based zone selection. This section evaluates the Rolling Horizon rollout layer, which wraps each base policy and overrides its drop decision whenever a short-horizon Monte Carlo evaluation identifies a better candidate. The rollout is evaluated on top of value-based zone selection for all four policies. Each rollout run uses the same seed as its corresponding base run, enabling paired cost comparisons.

### Results

Table 3 summarizes the rollout effect for each policy. The rollout provides no meaningful benefit for any of the four policies. Myopic's mean cost changes by only +€ 5 (+0.01 %) – negligibly worse with the rollout. For the three advanced policies the effect is equally small: Myopic+ saves € 19 (0.06 %), CFA saves € 64 (0.19 %), and DB saves € 44 (0.13 %). None of these differences exceed 0.2 % of mean total costs.

Table 3: Rollout effect per policy. All variants use value-based zone selection (500 runs each). Override rate = mean replan overrides / mean total carryover (≈48).

| Policy | Without Rollout (€) | With Rollout (€) | $\Delta$ (€) | $\Delta$ (%) | Ø Overrides | Override Rate |
|---|---|---|---|---|---|---|
| Myopic | 34,885 | 34,890 | $+5$ | $+0.01$ % | 0.78 | 1.6 % |
| Myopic+ | 34,313 | 34,294 | $-19$ | $-0.06$ % | 0.81 | 1.7 % |
| CFA | 33,793 | 33,729 | $-64$ | $-0.19$ % | 1.27 | 2.7 % |
| DB | 33,671 | 33,627 | $-44$ | $-0.13$ % | 1.20 | 2.5 % |

Figure 8 shows the paired cost difference (base minus rollout) per run for each policy. All four distributions are centered tightly on zero with roughly symmetric spread, confirming that the rollout neither systematically helps nor hurts any policy. Figure 9 shows the distribution of replan overrides per run. Across all four policies, most runs involve zero to two overrides; runs with more than four overrides are rare. CFA and DB trigger more overrides on average (1.20–1.27) than Myopic and Myopic+ (0.78–0.81). This is because CFA's and DB's value-weighted drop scores ($\tilde{C} -$ wage $\times$ detour) produce more differentiated candidate rankings, giving the Monte Carlo evaluation more opportunities to disagree; Myopic's uniform $1/$detour score results in more equally-ranked candidates where the rollout finds no clear alternative. Yet even the more frequently overriding policies show no meaningful cost improvement.

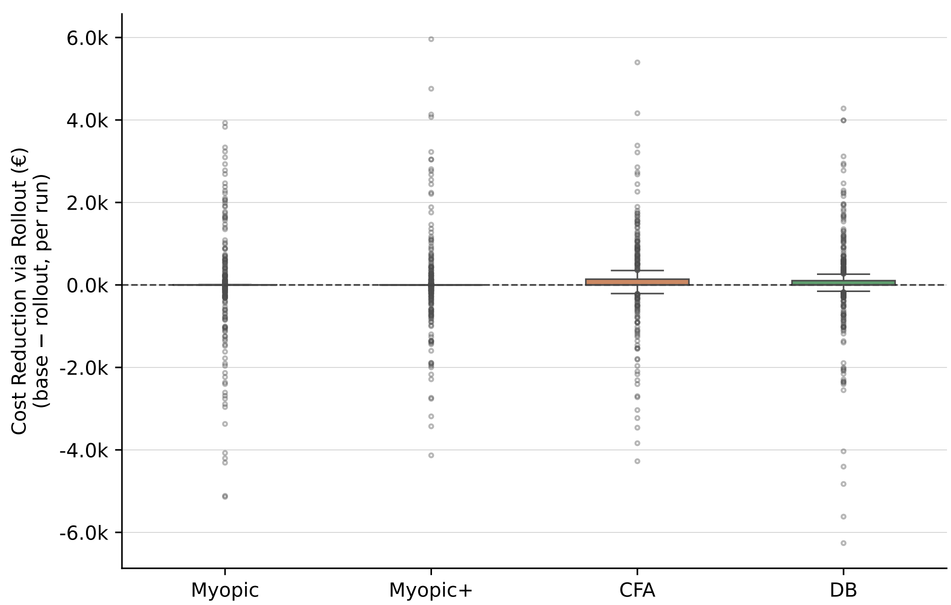


Figure 8: Paired cost reduction (base minus rollout) per run for each policy. Positive values indicate the rollout lowered cost. Dashed line marks zero. All variants use value-based zone selection (500 runs each).

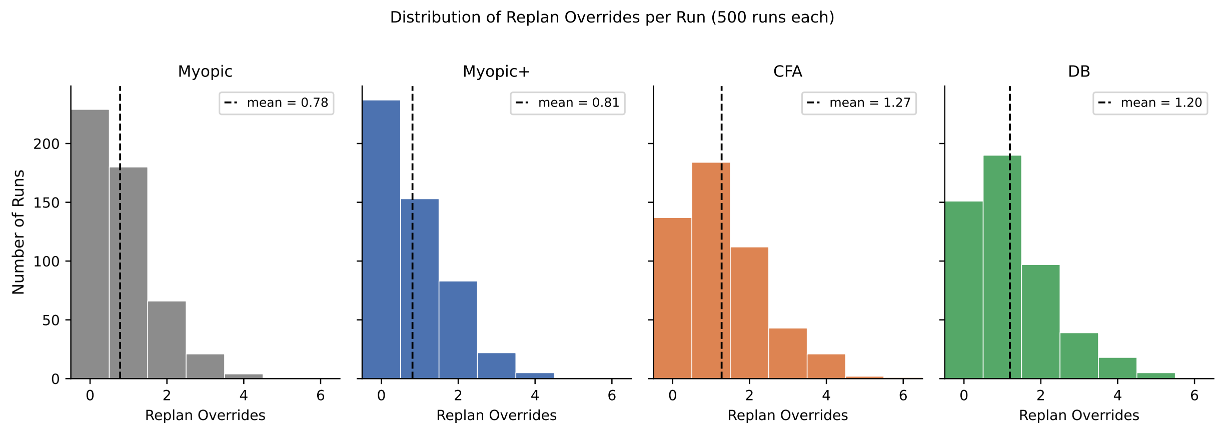


Figure 9: Distribution of replan overrides per run for all four rollout variants (500 runs each). Dashed vertical line shows the mean.

**Discussion**

The absence of any rollout benefit has a clear structural explanation. The rollout intervenes exclusively at the *drop decision*: when a new disruption cannot be inserted without dropping a routine task, the rollout evaluates the top-$k$ drop candidates via Monte Carlo lookahead and may override the base policy's choice. It does not affect how the initial daily plan is constructed.

For the advanced policies (Myopic+, CFA, DB), the rollout rarely disagrees with the base policy's drop choice (override rates of 1.2–1.3 per run), and when it does, the cost consequence is too small to accumulate into a measurable aggregate difference. Their drop-score functions already incorporate station value, so the Monte Carlo evaluation confirms rather than corrects their choices.

Myopic's situation is different in structure but leads to the same outcome. Myopic's drop-score uses $1/$detour – routing efficiency only, ignoring station power – and one might expect the rollout to systematically correct this. In practice, however, only 0.78 overrides per run occur, and they yield no net improvement. A contributing factor is that the rollout cannot address Myopic's initial-plan scoring function, which also ignores station power ($1/$distance versus Myopic+'s power$/$distance). Even if every drop decision were perfectly corrected, the ordering of routine stations within the initial plan would remain power-agnostic. The small remaining gap between Myopic and the advanced policies observed in Section 7.4 ($d = 0.17$–$0.37$) likely reflects this combined effect.

A practical implication follows: the rollout adds computational overhead without producing measurable cost improvements when paired with value-based zone selection. The dominant lever identified in Section 7.2 essentially saturates the available improvement potential before the rollout can contribute. Statistical validation of all findings in Sections 7.1–7.3 is provided in Section 7.4.

## 7.4 Statistical Validation

### Setup

All comparisons in Sections 7.1–7.3 are validated using paired statistical tests. Because each seed generates an identical stochastic disturbance sequence across all variants, runs sharing the same seed form natural pairs. This pairing eliminates the between-run variance due to the stochastic environment and substantially increases test power. Two complementary tests are applied: a paired $t$-test (parametric, assumes normally distributed differences) and a Wilcoxon signed-rank test (non-parametric, robust to non-normality and outliers). With $n = 500$ paired observations, even very small differences can yield statistically significant $p$-values. Cohen's $d$ (standardized mean difference of paired deltas) is therefore used as the primary criterion for practical relevance: $d < 0.20$ is considered negligible, $0.20$–$0.50$ small, $0.50$–$0.80$ medium, and $d > 0.80$ large.

### Results

Table 4 reports the key paired comparisons. Three distinct patterns emerge.

Table 4: Paired statistical tests for key comparisons (500 matched seeds each). Cohen's $d$ is the primary criterion for practical relevance; $p$-values are from the paired $t$-test and Wilcoxon signed-rank test.

| Comparison | Mean $\Delta$ (€) | $d$ | $t$-test $p$ | Wilcoxon $p$ | Finding |
|---|---|---|---|---|---|
| *Zone selection: centrality $\to$ value-based* | | | | | |
| Myopic | $-5{,}525$ | 1.29 | $< 0.001$ | $< 0.001$ | *** large |
| Myopic+ | $-5{,}831$ | 1.37 | $< 0.001$ | $< 0.001$ | *** large |
| CFA | $-6{,}394$ | 1.50 | $< 0.001$ | $< 0.001$ | *** large |
| DB | $-6{,}512$ | 1.49 | $< 0.001$ | $< 0.001$ | *** large |
| *Policy comparison under centrality* | | | | | |
| Myopic vs. Myopic+ | $+266$ | 0.07 | 0.128 | 0.137 | n.s. |
| Myopic vs. CFA | $+223$ | 0.06 | 0.181 | 0.157 | n.s. |
| Myopic vs. DB | $+227$ | 0.06 | 0.218 | 0.369 | n.s. |
| CFA vs. DB | $+4$ | 0.00 | 0.980 | 0.732 | n.s. |
| *Policy comparison under value-based zone selection* | | | | | |
| Myopic vs. DB | $+1{,}214$ | 0.37 | $< 0.001$ | $< 0.001$ | ** small |
| Myopic vs. CFA | $+1{,}092$ | 0.31 | $< 0.001$ | $< 0.001$ | ** small |
| Myopic vs. Myopic+ | $+573$ | 0.17 | $< 0.001$ | $< 0.001$ | * negligible |
| Myopic+ vs. DB | $+642$ | 0.20 | $< 0.001$ | $< 0.001$ | * negligible |
| Myopic+ vs. CFA | $+520$ | 0.16 | $< 0.001$ | 0.002 | * negligible |
| CFA vs. DB | $+122$ | 0.05 | 0.258 | 0.568 | n.s. |
| *Rollout effect (value-based base $\to$ +Rollout)* | | | | | |
| Myopic | $+5$ | 0.01 | 0.900 | 0.310 | n.s. |
| Myopic+ | $-19$ | 0.02 | 0.590 | 0.626 | n.s. |
| CFA | $-64$ | 0.08 | 0.073 | $< 0.001$ | negligible† |
| DB | $-44$ | 0.05 | 0.268 | $< 0.001$ | negligible† |

† Wilcoxon detects a statistically significant but practically negligible systematic shift ($d < 0.10$).

**Zone selection (d = 1.29–1.50):** The strongest effect in the entire study. For all four policies, switching from centrality to value-based zone selection produces differences of 1.3–1.5 standard deviations, significant at $p < 0.001$ under both tests. This is the only effect with a large Cohen's $d$.

**Policy differences under centrality (d = 0.00–0.07):** All comparisons are statistically non-significant ($p > 0.10$) with negligible effect sizes. No policy outperforms any other in a practically meaningful way when zone selection is held constant at centrality.

**Policy differences under value-based zone selection (d = 0.05–0.37):** Myopic differs significantly from all three advanced policies (d = 0.17–0.37, $p < 0.001$). Myopic+ also differs significantly from CFA and DB (d = 0.16–0.20, $p < 0.001$), though the effect sizes are negligible by Cohen's convention. The one non-significant comparison is CFA vs. DB (d = 0.05,

$p = 0.26$), confirming these two policies are effectively equivalent once zone selection is optimized.

**Rollout (d = 0.01–0.08):** No rollout comparison exceeds $d = 0.10$. For Myopic and Myopic+ the $t$-test and Wilcoxon both return non-significant results. For CFA and DB, the Wilcoxon detects a statistically significant but practically negligible systematic shift ($d = 0.08$ and $d = 0.05$ respectively) – a consequence of the large sample size ($n = 500$) rather than an economically meaningful effect.

Figure 10 summarizes all comparisons by Cohen's $d$. The zone-selection effects stand apart clearly; all other effects fall below the $d = 0.50$ threshold, and most are below $d = 0.20$.

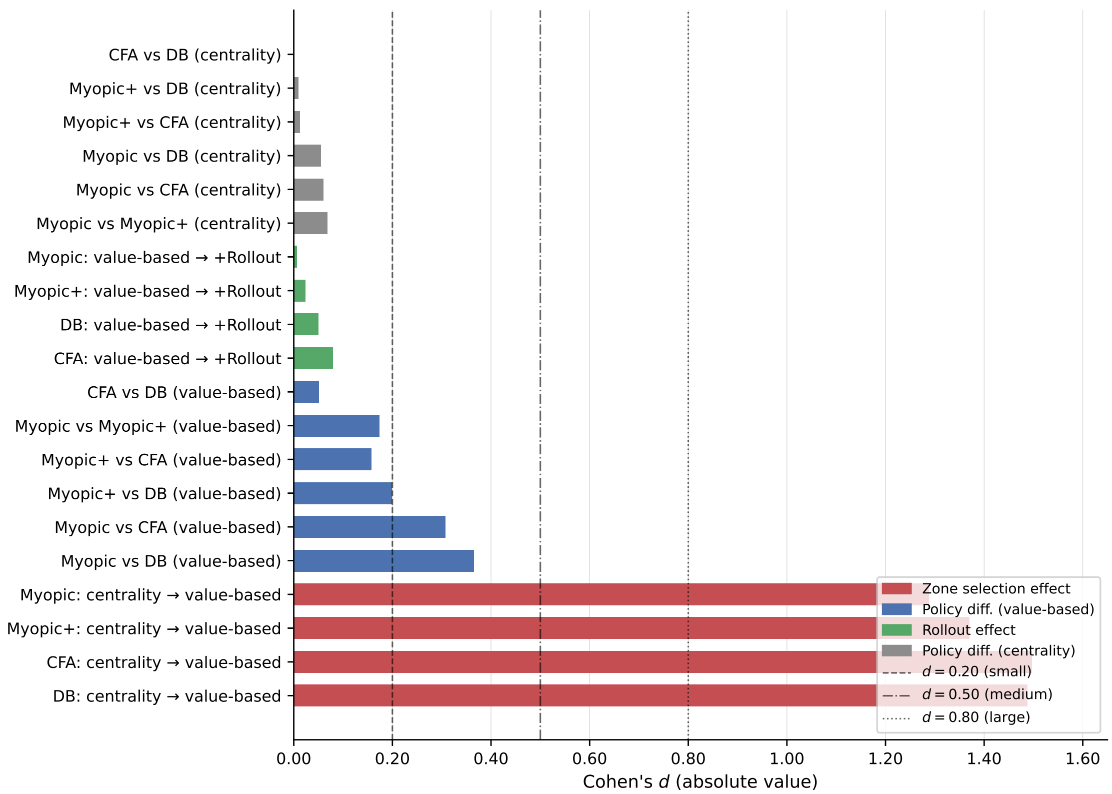


Figure 10: Cohen's $d$ for all key comparisons. Dashed vertical lines mark conventional thresholds (small: 0.20, medium: 0.50, large: 0.80). Zone-selection effects are the only comparisons with a large effect size; all other comparisons fall below $d = 0.40$.

### Discussion

The statistical analysis confirms and sharpens the findings from Sections 7.1–7.3. Three conclusions stand out.

First, zone selection is the only intervention with a large and robust effect. The $d > 1.2$ values for all four policies place this effect firmly in the "large" category by any standard, and the consistency across policies (all four show similar magnitudes) rules out policy-specific confounders. With $n = 500$ matched pairs, the $p$-values are effectively zero, and both parametric and non-parametric tests agree.

Third, the absence of a meaningful rollout effect is confirmed statistically, not just descriptively. For Myopic and Myopic+, neither test detects any systematic improvement. The Wilcoxon does flag CFA and DB rollouts as statistically significant – but with $d < 0.10$, this reflects the sensitivity of large-sample tests to trivially small effects, not a practically relevant finding. Given the computational cost of Monte Carlo rollouts at every disruption event, the rollout layer does not justify its overhead in this setting.

Second, under value-based zone selection, policy differences become detectable though remain small. Myopic differs significantly from all three advanced policies ($d = 0.17$–$0.37$), and Myopic+ differs from CFA and DB ($d = 0.16$–$0.20$), even though all four policies were statistically indistinguishable under centrality. The residual gap reflects the within-zone scoring logic: Myopic uses $1/$distance and $1/$detour (power-agnostic), while Myopic+, CFA, and DB weight decisions by station value. Once zone selection directs teams to high-value zones, these within-zone differences become visible – but given effect sizes between 0.16 and 0.37, the practical magnitude is limited.

# 8 Discussion

## 8.1 Interpretation of Results

The experimental results admit a clear, hierarchical reading. Three effects are present in the data, and they differ substantially in magnitude.

Switching from centrality-based to value-based zone selection reduces mean total costs by 13.7–16.2 % across all four policies, corresponding to Cohen's $d$ values of 1.29–1.50. This is the only effect in the study that qualifies as large by conventional standards, and it is fully consistent across policies and scenarios. The mechanism is straightforward: centrality-based selection steers teams toward zones that are geographically convenient but may contain stations with low economic urgency. Value-based selection directs teams toward zones where the expected downtime cost avoided is highest. Because downtime costs depend jointly on rated station power and days since last maintenance, a pure distance proxy systematically misallocates service capacity. The 29–34 % reduction in downtime costs observed across all policies confirms that the zone assignment itself, rather than the routing logic within zones, is the primary cost driver in this problem.

Policy differences are secondary and context-dependent. Under centrality-based zone selection, all four policies perform within 0.6 % of each other and no pairwise comparison is statistically significant ($d = 0.00$–$0.07$). The reason is structural: when zone selection does not differentiate between economically relevant and irrelevant stations, the value-weighted scoring functions of Myopic+, CFA, and DB have no meaningful signal to act on. The four policies share the same greedy cheapest-insertion backbone and differ only in how they score routine tasks and drop candidates. If the assigned station pool is economically homogeneous, different orderings produce similar aggregate costs.

Under value-based zone selection, policy differences become statistically detectable but remain practically limited. Myopic differs from the three advanced policies by $d = 0.17$–$0.37$; Myopic+ differs from CFA and DB by $d = 0.16$–$0.20$. The residual gap reflects Myopic's power-agnostic scoring ($1/$distance in the initial plan, $1/$detour in replanning): once high-value zones are assigned, routing within those zones in a value-blind manner forgoes a small but statistically detectable improvement. Notably, CFA and DB are statistically indistinguishable ($d = 0.05$, $p = 0.26$): the additional flexibility introduced by DB's distance-weighting parameter $\delta$ provides no measurable benefit over CFA's fixed formulation in the current experimental setting.

The Rolling Horizon layer overrides the base policy's drop choice in at most 1.3 events per run on average, and the resulting cost changes ($d = 0.01$–$0.08$) are economically negligible. The Wilcoxon test detects a statistically significant directional shift for CFA and DB, but with $d < 0.10$ and $n = 500$ observations, this reflects the sensitivity of large-sample tests to trivially small systematic differences rather than a practically relevant finding. The structural explanation is that the base policies' drop-score functions – especially the value-weighted ones – already incorporate the economic information that the Monte Carlo roll-out is designed to capture. When the base policy's greedy choice is already well-calibrated, lookahead rarely disagrees, and when it does, the cost difference is small.

Taken together, the results point to a clear priority ordering for practitioners: invest in zone-selection quality first, routing policy sophistication second. A well-chosen zone assignment with the simplest policy (Myopic) outperforms a poorly chosen zone assignment with the most sophisticated policy (DB) by a wide margin.

## 8.2 Limitations

The results reported in this thesis rest on a set of modelling and data assumptions that bound both their validity and their generalizability. This section identifies the most consequential limitations.

The entire experimental design assumes one central depot in Würzburg and 397 spatially concentrated charging stations. All routing and clustering logic is calibrated to this geometry. Operators with multiple depots, distributed vehicle fleets, or spatially sparse networks would face a substantially different problem structure. The K-Means zone partitioning and the greedy cheapest-insertion backbone are not inherently limited to single-depot settings, but the zone-selection scoring – which rewards proximity to depot as a proxy for team reachability – would require rethinking in a multi-depot context.

The stochastic failure simulation uses a parametric model in which failure probability is a function of days since last maintenance and rated station power, governed by four scalar parameters ($p_1$, $p_2$, $\lambda$, and the initial recovery factor). These parameters were set to produce plausible failure rates rather than estimated from historical WVV records. An attempt to obtain operational data from WVV for calibration was made but did not receive a response. The quantitative cost figures reported in Chapter 7 therefore reflect the synthetic distribution rather than actual field conditions. Relative comparisons between policies remain valid under this assumption, since all policies face identical failure sequences, but absolute cost magnitudes should not be read as predictions of real operating costs.

The model fixes the number of maintenance teams at two and the daily service capacity at 20 stations per team. In practice, operators may adjust staffing levels, deploy temporary support teams during peak fault periods, or subcontract overflow tasks. None of these options are represented in the current model. Similarly, the model does not account for vehicle range constraints, spare part availability, or technician skill heterogeneity – all of which affect real-world maintenance planning.

Hourly traffic matrices are used to represent time-dependent travel conditions, computed from the OSRM routing engine using OpenStreetMap data. While this is a reasonable approximation, OSRM does not model real-time congestion, road closures, or seasonal variation. The routing component therefore represents a simplified but not unrealistic version of actual travel conditions in Würzburg.

The rollout layer evaluates the top-$k$ drop candidates via Monte Carlo simulation at each disruption event. In the current setting ($k = 3$, 30-day horizon, 5 scenarios per candidate), this adds modest computational overhead. However, the cost scales linearly with the number of disruptions per day, the number of candidates $k$, and the simulation depth. For operators with larger fleets, higher disruption rates, or tighter replanning time budgets, the overhead could become prohibitive. The negligible performance benefit observed here makes this a moot point for the current problem instance, but scalability remains a concern if the approach is extended to more dynamic settings.

The optimization target is a scalar cost composed of labor, travel, and downtime cost components. In practice, maintenance operators face multi-objective trade-offs: service level agreements may impose hard response-time constraints, equity considerations may require balanced workloads across teams, and customer satisfaction may be a metric distinct from pure economic downtime cost. The model does not represent these objectives, and results may differ under alternative objective specifications.

## 8.3 Practical Relevance and Transferability

The thesis was motivated by a concrete operational problem: how should a maintenance team for a growing EV charging network plan and adapt its daily routes under uncertainty? The Würzburg charging network operated by the Würzburg Versorgungs- und Verkehrs-GmbH (WVV) served as the reference case. An attempt to obtain operational data and benchmark figures directly from WVV was made during the preparation of this thesis but received no response. A direct empirical comparison between the modelled policies and the operator's current scheduling practice is therefore not possible. The following discussion addresses the practical relevance of the findings at a conceptual level and considers the conditions under which the approach could be deployed or transferred to other settings.

Deploying the system in a real maintenance operation would require three types of input data that are not always readily available. Historical failure records per station are needed to calibrate the stochastic failure model – specifically to estimate per-station failure rates and the relationship between maintenance recency and subsequent failure probability. Station-level metadata (rated power, installation date, precise GPS coordinates) must be available in structured form. Finally, real-time fault notifications must be routable to the planning system, either via sensor telemetry or a fault-reporting interface. None of these requirements are technically prohibitive for an operator of WVV's scale, but they imply a data management infrastructure that may not exist in its current form.

The approach is best suited to operators with a centralized depot structure, a stable fleet of two to four maintenance teams, and a medium-density station network in a single metropolitan area. These conditions match the WVV setting and also describe many municipal or regional charging network operators in Germany. The core computational cost is low: the greedy routing policies run in milliseconds, and even the zone-selection logic adds negligible overhead. The only computationally intensive component – the Rolling Horizon rollout – was shown to provide no meaningful benefit in this setting and can be omitted without loss.

Three structural features of the current model constrain transferability to other contexts. The zone-selection scoring function assumes a single depot; operators with spatially distributed vehicle pools would need to replace the depot-distance term with a fleet-assignment-aware proximity measure. The CFA cost approximation $\tilde{C}$ was trained on synthetic data from the Würzburg network, and its learned weights $\theta$ are unlikely to transfer directly to networks with substantially different station densities, power distributions, or failure dynamics. Retraining on local data is straightforward given the offline regression procedure, but requires sufficient historical observations. The model also assumes homogeneous service times; operators with a mix of hardware generations or station types requiring different maintenance procedures would need to extend the time-window logic accordingly.

Despite these constraints, the central finding – that zone selection quality dominates routing policy sophistication – is likely to transfer beyond the specific experimental setting. It implies that a maintenance operator's most impactful intervention is not algorithmic refinement of the routing heuristic, but rather the development of a zone-value signal that incorporates station power and maintenance recency. This information is typically available to operators from internal maintenance records alone. A simple rule-based implementation of value-based zone selection, without any of the ADP machinery developed in this thesis, could plausibly capture a substantial fraction of the cost improvement observed here. The more sophisticated policies (CFA, DB) provide incremental benefits that are statistically detectable but economically modest – on the order of 3–4 % relative to a well-configured Myopic+ baseline under value-based zone selection.

# 9 Outlook and Future Work

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# A Anhang A

Table 5: Summary of notation used in Chapter 4

| Symbol | Description |
|---|---|
| *Sets and indices* | |
| $I = \{1, \ldots, 397\}$ | Set of charging stations |
| $K = \{1, 2\}$ | Set of service teams |
| $V = \{0\} \cup I$ | Complete node set (depot + stations) |
| $Z = \{1, \ldots, N_Z\}$, $N_Z = 178$ | Set of zones |
| $H = \{8, 9, \ldots, 16\}$ | Hourly decision grid |
| $i, j \in V$ | Node indices |
| $k \in K$ | Team index |
| $h \in H$ | Decision epoch |
| $d_{ij}$ | Travel time from node $i$ to node $j$ [min] |
| *State components* | |
| $s_h = (\tau_h,\, X_h,\, Y_h,\, J_h,\, \Pi_h)$ | System state at epoch $h$ |
| $\tau_h$ | Elapsed workday time [min] |
| $X_h$ | Team states at epoch $h$ |
| $v_h^k$ | Current location of team $k$ |
| $\sigma_h^k$ | Activity status of team $k$ |
| $b_h^k$ | Remaining time budget of team $k$ [min] |
| $T_{\max}$ | Total working day length [min] |
| $Y_h$ | Station maintenance states at epoch $h$ |
| $m_h^i \in \{0, 1\}$ | Periodic maintenance completion indicator |
| $dsm_h^i$ | Days since last maintenance of station $i$ |
| $J_h = (W_h^{\text{reg}},\, W_h^{\text{fault}},\, W_h^{\text{carry}})$ | Open maintenance jobs |
| $W_h^{\text{reg}}$ | Regular maintenance jobs |
| $W_h^{\text{fault}}$ | Newly arrived fault jobs |
| $W_h^{\text{carry}}$ | Carryover fault jobs from previous day |
| $\Pi_h$ | Planned routes for both teams |
| *Actions and dynamics* | |
| $A(s_h)$ | Feasible action set in state $s_h$ |
| $a_h \in A(s_h)$ | Chosen action at epoch $h$ |
| $P(s_{h+1} \mid s_h, a_h)$ | Transition function |
| *Costs and value* | |
| $R = -(C_{\text{op}} + C_{\text{dt}})$ | Reward function |
| $C_{\text{op}}$ | Operational cost (labor + travel) |
| $C_{\text{dt}}$ | Downtime cost |
| $n_{\text{active}}$ | Number of active teams on a given day |

Table 5 – continued

| **Symbol** | **Description** |
|---|---|
| $T_{\mathrm{shift}}$ | Effective shift duration [h] |
| $\mathrm{km}_{ij}$ | Distance from node $i$ to node $j$ [km] |
| $p_j$ | Nominal charging power of station $j$ [kW] |
| $\Delta\tau_j^{\mathrm{wait}}$ | Effective waiting time of fault $j$ [h] |
| $c_L = 35$ EUR/h | Labor cost rate |
| $c_F = 0.30$ EUR/km | Travel cost rate |
| $c_{\mathrm{dt}} = 0.50$ EUR/kWh | Downtime cost rate |
| $Q^*(s, a)$ | Optimal state-action value function |
| *Fault model* | |
| $p_{\ell,\mathrm{base}}$ | Baseline fault rate for fault type $\ell$ [per hour] |
| $\eta^i$ | Station-level fault scaling factor |
| $f(t)$ | Recovery curve |
| $\lambda$ | Residual fault factor immediately after maintenance |
| $\tau_{\mathrm{rec}}$ | Recovery period [days] |

Hiermit versichere ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie die Zitate deutlich kenntlich gemacht zu haben.
Ich erkläre weiterhin, dass die vorliegende Arbeit in gleicher oder ähnlicher Form noch nicht im Rahmen eines anderen Prüfungsverfahrens eingereicht wurde.

Würzburg, den 4. Juli 2026 VORNAME NACHNAME